BaseCore of the Rigid Identity Framework

Autonomous Mathematical Source Text for Foundational Dynamics, Identity Rigidity, Regime Constraints, and Operational-Criterion Grammar

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Abstract

This document is the BaseCore of the Rigid Identity Framework: the autonomous mathematical base layer of the source tree. It selectively normalizes formal material from Papers 1--6 only where theorem ownership belongs in the base layer. Papers 7--9 remain downstream and are not required for the validity of any theorem stated here. BaseCore proves contractive projection dynamics, typed computable witnesses, parameterwise non-collapse under explicit anti-constant assumptions, inverse-limit identity under non-finite-determination assumptions, conditional rigidity under admissible perturbation models, conditional obstruction to uniform response fidelity under a binary representation budget, operational-criterion grammar, and falsation / claim-boundary ledgers. It does not prove phenomenality, metaphysical subjectivity, human equivalence, runtime instantiation, bridge admissibility, or external adjudication. The LaTeX tree is the source of truth; the PDF is a compiled artifact only.

Contents

I Foundational Dynamics **5**

1 Scope, System Boundary, and Non-Inference Note **5**

2 Introduction **5**
2.1 Purpose 5
2.2 Methodology 5
2.3 Scope discipline 5

3 Minimal Base Hypotheses **5**

4 The Projection Operator **6**

5 The Transition Operator **7**

6 Existence and Compactness of Attractors **7**

7 Parameterwise Non-Collapsibility **7**

8 Summary of Base Results **8**

II Typed Computable Model and State Spectral Gap **8**

9 Typed Computable Model: Embedded Constant-Disk Witness **8**
9.1 Golden Mean Parameter Space . . . 9
9.2 Typed Banach Witness . . . 9
9.3 Basic Operator Properties . . . 10
9.4 Separate Anti-Collapse Witness . . . 11

10 State Spectral Gap and Parameter-Family Persistence **12**

III Identity, Rigidity, and Conditional Non-Simulability **13**

11 Projective Systems and Inverse-Limit Identity **13**
11.1 Projective Setup . . . 13
11.2 Non-Locality of Identity . . . 13

12 Rigidity under Admissible Perturbations **14**
12.1 Metric Projective Assumptions . . . 14

13 Conditional Non-Simulability **16**

14 Section Summary **18**

IV Regime Structure and Continuity Constraints **18**

15 Formal Setup **18**
15.1 Review of Identity Framework . . . 18
15.2 Abstract Phenomenological Space . . . 18
15.2.1 Why a Separate Space? . . . 18
15.3 Phenomenological Assignment . . . 19
15.3.1 Uniqueness of the Limit Assignment . . . 19
15.4 Axioms for Phenomenological Coherence . . . 19
15.5 Status of the Assignment Conditions . . . 20
15.5.1 Nontriviality of the Codomain . . . 20
15.5.2 Controlled Convergence . . . 20
15.5.3 Finite-Cut Non-Factorization . . . 20
15.6 Logical Dependencies and Realizability . . . 20
15.7 Rejected Alternatives . . . 20
15.7.1 Rejected: Continuity Axiom . . . 21
15.7.2 Rejected: Surjectivity Axiom . . . 21
15.7.3 Rejected: Injectivity Axiom . . . 21
15.8 Structural Interpretation . . . 21
15.9 Closure of Section 2 . . . 21
15.10Phenomenological Neutrality Clause . . . 21
15.11Exact Compatibility Obstruction . . . 22
15.12A Nonempty Controlled Assignment Model . . . 22
15.13Common Readout on Two Coherently Transported Towers . . . 23
15.14Controls Separating Convergence, Continuity, and Invariance . . . 24

16 Compatibility with Identity Regimes **24**
16.1 Induced Constraints from Identity Rigidity 24
16.2 Metric Structure for Quantitative Comparisons 24
16.2.1 Role of the Metric 25
16.2.2 Completeness as a Limit Guarantee 25
16.2.3 Alternatives Considered and Rejected 25
16.3 Phenomenological Continuity 25
16.4 Phenomenological Fragmentation 26
16.5 Relationship Between Metrics 26
16.6 Closure of Section 3 26

17 Impossibility Theorems **26**
17.1 Φ-Regularity Hypothesis 26
17.2 Fragmentation Theorem 27
17.3 Forced Continuity Theorem 27
17.4 Dichotomy Theorem 28

18 Classification of Phenomenological Regimes **28**
18.1 Conditional Classification 28
18.2 Structural Implications 28

19 Extended Proofs **29**
19.1 Proof of Fragmentation Theorem (Extended) 29
19.2 Proof of Forced Continuity Theorem (Extended) 29
19.3 Structural Necessity Lemma 29

V Null-Regime Structural Exclusion **29**

20 Compatibility, Witnesses, and Non-Nullity **29**
20.1 Compatibility Operator 29
20.2 Witness-Relative Null Separation 30
20.3 Positive Regime Structure 31

21 Canonical Families and Structural Closure Consequences **32**

VI BaseCore Boundary and Expansion Interface **32**

22 BaseCore Ownership and Downstream Boundary **33**

23 Admissible Structural Extensions **33**

VII Operational-Criterion Grammar and Certification **34**

24 Admissible Systems, Supports, and Histories **35**
24.1 System Model 35
24.2 Admissible Support 35
24.3 Operational Histories 36
24.4 Witness Margin Vector 36

25 Six Critical Invariants **36**
25.1 Structural Persistence . . . 36
25.2 Rigid Identity . . . 37
25.3 Causal Integration . . . 37
25.4 Regime Continuity . . . 37
25.5 Non-Null Differentiation . . . 38
25.6 Operational Legibility . . . 38

26 Criterion Class, Certified Partition, and Rupture **39**

27 Minimality of the Invariant Basis **40**

28 Operational Legibility as a Certified Object **40**

29 Certified Candidate Audit Schema **41**

VIII Claim Boundary and Falsation Grammar **41**

30 Forensic Admissibility and Integrity Grammar **41**

31 Certification, Comparison, and Rupture **42**

32 Prediction and Falsation Grammar **42**

33 Claim Boundary Ledger **43**
33.1 Allowed BaseCore Claims . . . 43
33.2 Forbidden BaseCore Claims . . . 43

34 Dependency Ledger **43**

35 Assumption-to-Theorem-to-Non-Claim Mapping **44**

36 Theorem Hygiene Ledger **45**

37 Redundancy Ledger **45**

38 Open Assumptions Inventory **46**

39 Downstream Boundary Note **46**

IX Discrete-to-Continuous Bridge **46**

40 From Node Networks to Hilbert Space Dynamics **46**

41 Projective Dissipation on the Circle-Prefix Towers **50**

A Counterexamples for Necessity **52**
A.1 Without H1 (Convexity) . . . 52
A.2 Without H2 ($\|K\| < 1$) . . . 53
A.3 Without H3 (Completeness) . . . 53
A.4 Without H4 (Compactness) . . . 53
A.5 Why Φ-Regularity Is a Hypothesis . . . 53

Part I
Foundational Dynamics

1 Scope, System Boundary, and Non-Inference Note

What BaseCore does. This document proves the mathematical base results that later layers may use: projection existence and uniqueness, strict contraction, unique fixed points, compact attractor families, parameterwise non-collapse under explicit anti-constant assumptions, inverse-limit identity under non-finite-determination assumptions, conditional rigidity, conditional non-simulability, and operational-criterion grammar.
What BaseCore is not. BaseCore is not the whole QICN corpus. It is not a bridge program, not runtime validation, not a phenomenality theory, not a human-equivalence claim, and not an external-adjudication layer.
What should not be inferred. Nothing in BaseCore alone licenses claims about consciousness, subjectivity, phenomenality, moral status, human-machine equivalence, bridge admissibility, or deployment readiness. Those are downstream burdens, not consequences of the theorems below.
Source-of-truth rule. The LaTeX tree is the source of truth. Compiled PDFs, release bundles, and runtime artifacts are derived surfaces only.

2 Introduction

2.1 Purpose

BaseCore is the canonical base layer of the Rigid Identity Framework. Its role is narrower and stronger than a broad architectural survey: it owns only the theorem-level mathematics that downstream papers may depend on without making BaseCore depend on them.

2.2 Methodology

We use standard tools from:

- Hilbert-space geometry and metric projections;
- contraction mappings and fixed-point theory;
- compactness and continuity arguments;
- projective systems and inverse limits;
- assumption-led claim-boundary discipline.

2.3 Scope discipline

This section is pure mathematics. The symbols ‘‘identity,’’ ‘‘regime,’’ and ‘‘criterion’’ are structural labels only. Their ordinary-language interpretations are not part of any proof unless explicitly formalized as assumptions.

3 Minimal Base Hypotheses

Hypothesis 3.1 (H1: Projection Structure)**.** Let $\mathcal{H}$ be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. Let $\mathcal{I} \subset \mathcal{H}$ be non-empty, closed, and convex.

Hypothesis 3.2 (H2: Contraction Operator)**.** Let $K : \mathcal{H} \to \mathcal{H}$ be a bounded linear operator with operator norm $\|K\| < 1$.

Hypothesis 3.3 (H3: Completeness)**.** The ambient space $\mathcal{H}$ is complete.

Remark 3.4 (Logical Status of Completeness)*.* Completeness is already part of the Hilbert-space assumption in H1. H3 records this requirement explicitly for later generalizations; it is not independent of H1 as written. A counterexample in an incomplete inner-product space tests such a generalization, not independence of H3 from H1.

Hypothesis 3.5 (H4: Parameter Compactness)**.** Let $\mathcal{U}$ be a compact metric space and let $\Gamma : \mathcal{U} \to \mathcal{H}$ be uniformly continuous.

Proposition 3.6 (Base sufficiency of H1--H4)**.** *Hypotheses H1--H4 are sufficient for:*

1. existence and uniqueness of the metric projection onto $\mathcal{I}$;

2. strict contraction of the transition operators T_u;

3. existence and uniqueness of the fixed point f_u^* for each $u \in \mathcal{U}$;

4. compactness of the attractor family $\mathcal{A}^* = \{f_u^* : u \in \mathcal{U}\}$.

They do not, by themselves, imply parameterwise non-collapse.

Proof. The four items are exactly the content of Theorems 4.2, 5.2, 6.1, and 6.3. Parameterwise non-collapse requires an additional anti-constant hypothesis introduced in Section 7. □

4 The Projection Operator

Definition 4.1 (Metric Projection)**.** For $\Psi \in \mathcal{H}$, define the metric projection onto $\mathcal{I}$ by

$$\aleph(\Psi) := \arg\min_{\Phi \in \mathcal{I}} \|\Psi - \Phi\| .$$

Theorem 4.2 (Existence and Uniqueness of Projection)**.** *Under Hypothesis 3.1, the projection $\aleph(\Psi)$ exists and is unique for every $\Psi \in \mathcal{H}$.*

Proof. Existence follows because a minimizing sequence in the closed convex set $\mathcal{I}$ is Cauchy by the parallelogram law, hence converges in the complete space $\mathcal{H}$ to a point of $\mathcal{I}$ that attains the infimum. Uniqueness follows from strict convexity of the norm induced by the Hilbert inner product: if two minimizers existed, their midpoint would produce the same minimum and the parallelogram identity would force the two minimizers to coincide. □

Lemma 4.3 (Non-Expansiveness of the Projection)**.** *The map $\aleph : \mathcal{H} \to \mathcal{I}$ is non-expansive:*

$$\|\aleph(\Psi_1) - \aleph(\Psi_2)\| \le \|\Psi_1 - \Psi_2\| \qquad \forall \Psi_1, \Psi_2 \in \mathcal{H}.$$

Proof. This is the standard Hilbert-space projection inequality. Let $\Phi_i = \aleph(\Psi_i)$. The projection characterization

$$\langle \Psi_i - \Phi_i, \Phi - \Phi_i \rangle \le 0 \qquad \forall \Phi \in \mathcal{I}$$

applied with $\Phi = \Phi_2$ and $\Phi = \Phi_1$ yields

$$\|\Phi_1 - \Phi_2\|^2 \le \langle \Psi_1 - \Psi_2, \Phi_1 - \Phi_2 \rangle \le \|\Psi_1 - \Psi_2\| \, \|\Phi_1 - \Phi_2\| .$$

If $\Phi_1 \neq \Phi_2$, divide by $\|\Phi_1 - \Phi_2\|$. □

5 The Transition Operator

Definition 5.1 (Transition Operator)**.** For each $u \in \mathcal{U}$, define

$$T_u(\Psi) := \aleph(K\Psi + \Gamma(u)).$$

Theorem 5.2 (Contractivity)**.** *Under Hypotheses 3.1 and 3.2, each T_u is a strict contraction with Lipschitz constant at most $\|K\|$:*

$$\|T_u(\Psi_1) - T_u(\Psi_2)\| \leq \|K\| \, \|\Psi_1 - \Psi_2\| .$$

Proof. By Lemma 4.3,

$$\begin{aligned} \|T_u(\Psi_1) - T_u(\Psi_2)\| &= \|\aleph(K\Psi_1 + \Gamma(u)) - \aleph(K\Psi_2 + \Gamma(u))\| \\ &\leq \|K(\Psi_1 - \Psi_2)\| \leq \|K\| \, \|\Psi_1 - \Psi_2\| . \end{aligned}$$

Since $\|K\| < 1$, strict contraction follows. □

6 Existence and Compactness of Attractors

Theorem 6.1 (Unique Fixed Point)**.** *Under Hypotheses 3.1--3.3, for each $u \in \mathcal{U}$ there exists a unique $f_u^* \in \mathcal{H}$ such that*

$$T_u(f_u^*) = f_u^*.$$

Moreover, for every $\Psi_0 \in \mathcal{H}$,

$$\lim_{n\to\infty} T_u^n(\Psi_0) = f_u^*.$$

Proof. By Theorem 5.2, each T_u is a strict contraction on the complete metric space $(\mathcal{H}, \|\cdot\|)$. The Banach fixed-point theorem yields existence, uniqueness, and convergence of iterates. □

Definition 6.2 (Attractor Family)**.** Define the attractor family

$$\mathcal{A}^* := \{f_u^* : u \in \mathcal{U}\}.$$

Theorem 6.3 (Compactness of the Attractor Family)**.** *Under Hypotheses 3.1--3.5, the set $\mathcal{A}^*$ is compact in $\mathcal{H}$.*

Proof. Let $F : \mathcal{U} \to \mathcal{H}$ be given by $F(u) = f_u^*$. Insert the two fixed-point equations and use non-expansiveness of projection to obtain

$$\|f_u^* - f_v^*\| \leq \|K\| \, \|f_u^* - f_v^*\| + \|\Gamma(u) - \Gamma(v)\| .$$

Consequently

$$\|f_u^* - f_v^*\| \leq \frac{\|\Gamma(u) - \Gamma(v)\|}{1 - \|K\|}.$$

Thus F is continuous. Since $\mathcal{U}$ is compact and $\mathcal{A}^* = F(\mathcal{U})$, the image is compact. □

7 Parameterwise Non-Collapsibility

Definition 7.1 (Constant Subspace)**.** Whenever the ambient realization distinguishes constant elements, let $\mathcal{N} \subset \mathcal{H}$ denote the closed subspace of constants.

Definition 7.2 (Parameterwise Non-Collapsible Family)**.** A family $\{T_u\}_{u\in\mathcal{U}}$ is parameterwise non-collapsible if

$$f_u^* \notin \mathcal{N} \qquad \forall u \in \mathcal{U}.$$

Hypothesis 7.3 (H5: No Constant Fixed Points)**.** For every parameter $u \in \mathcal{U}$ and every constant element $c \in \mathcal{N}$,

$$T_u(c) \neq c.$$

Theorem 7.4 (Structural Non-Collapsibility)**.** *Under Hypotheses 3.1--7.3, the family is parameterwise non-collapsible:*

$$f_u^* \notin \mathcal{N} \qquad \forall u \in \mathcal{U}.$$

Proof. Fix $u \in \mathcal{U}$. Suppose by contradiction that $f_u^* \in \mathcal{N}$. Write $f_u^* = c$ with $c \in \mathcal{N}$. Since f_u^* is a fixed point,

$$T_u(c) = T_u(f_u^*) = f_u^* = c,$$

contradicting Hypothesis 7.3. Therefore $f_u^* \notin \mathcal{N}$. □

Remark 7.5 (What H5 does not say). Hypothesis 7.3 is parameterwise. It does not merely say that *some* parameter breaks constant collapse. It excludes constant fixed points for *every* parameter. This is exactly the quantifier strength needed for Theorem 7.4.

Proposition 7.6 (H5 Is Not Implied by H1--H4)**.** *There exists a family satisfying H1--H4 for which H5 fails and its unique equilibrium is constant. There also exists a family satisfying H1--H4 for which H5 holds with a positive uniform distance of the equilibria from the constant subspace.*

Proof. For the first assertion, take $\mathcal{H} = L^2([0,1])$, $\mathcal{I} = \mathcal{H}$, $K = q\,\mathrm{Id}$ with $0 \leq q < 1$, $\mathcal{U} = \{0\}$ and $\Gamma(0) = 0$. Projection is the identity, $T_0 f = qf$, and the unique fixed point is the constant zero function. H1--H4 hold, but H5 fails at $c = 0$.

For the second assertion, Theorem 40.15 and Example 40.16 construct a family with $\mathcal{I} = \{0 \leq f \leq 1\}$, $Kf = \frac{1}{4}\int f$ and $\Gamma(u)(s) = s/4 + u$, $u \in [0, 1/4]$. Its explicitly computed unique fixed points have distance $1/(4\sqrt{12})$ from the constants for every parameter. The proof uses the displayed dynamics and does not assume H5 to derive that distance. Thus H1--H4 permit both regimes; a structural condition beyond them is needed to select non-collapse. □

8 Summary of Base Results

1. H1--H4 give a complete theorem-tight base for projection existence, strict contraction, unique fixed points, and compact attractor families.
2. H5 is a separate anti-constant assumption block used only for parameterwise non-collapse.
3. No empirical, phenomenological, biological, or runtime claim is inferred at this level.
4. Later sections may add conditional identity, rigidity, and criterion theorems, but only by introducing their own explicit assumption blocks.

Part II
Typed Computable Model and State Spectral Gap

9 Typed Computable Model: Embedded Constant-Disk Witness

This section repairs the type leaks of the previous computable model. The purpose is narrow: provide a fully typed witness for the transition grammar and the state spectral gap. It does *not* by itself certify parameterwise non-collapse; that burden is handled separately below.

9.1 Golden Mean Parameter Space

Definition 9.1 (Golden Mean Shift Parameter Space)**.** Let $\Sigma = \{0, 1\}$ and

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

The Golden Mean Shift $S_A \subset \Sigma^{\mathbb{N}}$ is the subshift of finite type consisting of all binary sequences that avoid the block 11.

Definition 9.2 (Parameter Encoding)**.** Define

$$u(x) = \sum_{i=1}^{\infty} x_i \phi^{-i}, \qquad \phi = \frac{1+\sqrt{5}}{2}.$$

The image $\mathcal{U} := u(S_A) \subset [0, 1]$ is compact.

9.2 Typed Banach Witness

Definition 9.3 (Ambient Space and Constant Embedding)**.** Let

$$X = [0, 1], \qquad E = \mathbb{R}^2, \qquad \mathcal{B} = C(X, E)$$

with norm

$$\|f\|_\infty := \sup_{s \in X} \|f(s)\|_2 .$$

Define the constant embedding

$$j : E \to \mathcal{B}, \qquad j(v)(s) = v.$$

Definition 9.4 (Invariant Constant Disk)**.** Let

$$D := \{v \in E : \|v\|_2 \le 1\}, \qquad \mathcal{I}_c := j(D) \subset \mathcal{B}.$$

Definition 9.5 (Averaging Operator)**.** Define

$$A : \mathcal{B} \to E, \qquad A(f) := \int_0^1 f(\tau)\, d\tau.$$

Definition 9.6 (Disk Projection)**.** Define the Euclidean projection

$$P_E : E \to D, \qquad P_E(v) = \begin{cases} v, & \|v\|_2 \le 1, \\ \dfrac{v}{\|v\|_2}, & \|v\|_2 > 1. \end{cases}$$

Definition 9.7 (Typed Projection onto the Constant Disk)**.** Define

$$\aleph_{\mathcal{B}} : \mathcal{B} \to \mathcal{I}_c, \qquad \aleph_{\mathcal{B}}(f) := j(P_E(A(f))).$$

Definition 9.8 (Constant-Valued Contraction Operator)**.** Define

$$\widetilde{K} : \mathcal{B} \to \mathcal{B}, \qquad (\widetilde{K} f)(s) := \frac{1}{2} \int_0^1 f(\tau)\, d\tau.$$

Definition 9.9 (Lifted Input Coupling)**.** Let $\Gamma : \mathcal{U} \to E$ be

$$\Gamma(u) = (u, 0).$$

Define its constant lift

$$\Gamma_{\mathcal{B}} : \mathcal{U} \to \mathcal{B}, \qquad \Gamma_{\mathcal{B}}(u) := j(\Gamma(u)).$$

Definition 9.10 (Typed Transition Operator)**.** For each $u \in \mathcal{U}$, define

$$T_u : \mathcal{B} \to \mathcal{I}_c, \qquad T_u(f) := \aleph_{\mathcal{B}}\big(\widetilde{K} f + \Gamma_{\mathcal{B}}(u)\big).$$

Remark 9.11 (Typing discipline). No unembedded element of E is treated as an element of $\mathcal{B}$. The embedding j is always written explicitly. This removes the earlier type leak in which a disk in E was silently treated as a subset of a function space.

9.3 Basic Operator Properties

Lemma 9.12 (The Embedding is Isometric)**.** *For every $v \in E$,*

$$\|j(v)\|_\infty = \|v\|_2 .$$

Proof. By definition,

$$\|j(v)\|_\infty = \sup_{s\in X} \|j(v)(s)\|_2 = \sup_{s\in X} \|v\|_2 = \|v\|_2 .$$

□

Lemma 9.13 (The Disk Projection is Non-Expansive)**.** *For all $v, w \in E$,*

$$\|P_E(v) - P_E(w)\|_2 \le \|v - w\|_2 .$$

Proof. P_E is the metric projection onto the closed convex set $D \subset \mathbb{R}^2$, so the standard Hilbert-space non-expansiveness property applies. □

Lemma 9.14 (The Typed Projection is Non-Expansive)**.** *For all $f, g \in \mathcal{B}$,*

$$\|\aleph_{\mathcal{B}}(f) - \aleph_{\mathcal{B}}(g)\|_\infty \le \|f - g\|_\infty .$$

Proof. Using the isometry of j, the non-expansiveness of P_E, and Jensen's inequality for the averaging operator,

$$\begin{aligned}
\|\aleph_{\mathcal{B}}(f) - \aleph_{\mathcal{B}}(g)\|_\infty &= \|P_E(A(f)) - P_E(A(g))\|_2 \\
&\le \|A(f) - A(g)\|_2 \\
&= \left\| \int_0^1 (f(\tau) - g(\tau))\, d\tau \right\|_2 \\
&\le \int_0^1 \|f(\tau) - g(\tau)\|_2 \, d\tau \le \|f - g\|_\infty .
\end{aligned}$$

□

Lemma 9.15 (Norm of the Lifted Contraction)**.**

$$\left\|\widetilde{K}\right\| \le \frac{1}{2}.$$

Proof. For any $f \in \mathcal{B}$,

$$\left\|\widetilde{K} f\right\|_\infty = \left\| \frac{1}{2} \int_0^1 f(\tau)\, d\tau \right\|_2 \le \frac{1}{2} \int_0^1 \|f(\tau)\|_2 \, d\tau \le \frac{1}{2} \|f\|_\infty .$$

□

Theorem 9.16 (Typed Contraction on the Computable Witness)**.** *For each $u \in \mathcal{U}$, the map $T_u : \mathcal{B} \to \mathcal{I}_c$ is a strict contraction:*

$$\|T_u(f) - T_u(g)\|_\infty \le \frac{1}{2} \|f - g\|_\infty .$$

Proof. Combine non-expansiveness of $\aleph_{\mathcal{B}}$ with the bound on $\widetilde{K}$:

$$\|T_u(f) - T_u(g)\|_\infty \le \left\|\widetilde{K} f - \widetilde{K} g\right\|_\infty \le \frac{1}{2} \|f - g\|_\infty .$$

□

Corollary 9.17 (Unique Fixed Point in the Typed Witness)**.** *For each $u \in \mathcal{U}$, there exists a unique $f_u^* \in \mathcal{I}_c$ such that*

$$T_u(f_u^*) = f_u^*.$$

Proof. The Banach fixed-point theorem applies on the complete metric space $(\mathcal{B}, \|\cdot\|_\infty)$. □

Proposition 9.18 (Typed Non-Vacuity of the Computable Model)**.** *The embedded constant-disk Golden Mean model is a fully typed, non-vacuous witness of the transition grammar:*

1. every operator has explicit domain and codomain;

2. the transition map T_u is well-defined on $\mathcal{B}$;

3. the family admits unique fixed points and a compact parameter image;

4. the state contraction bound is explicit and computable.

Proof. Items (1)--(3) follow from the definitions above and the contraction theorem. Compactness of the parameter image follows because $\mathcal{U}$ is compact and $u \mapsto f_u^*$ is continuous under a uniform contraction constant. □

Remark 9.19 (Why this witness does not certify H5). The image of T_u lies in the constant subspace $\mathcal{I}_c$. Therefore its fixed point is constant by construction. This witness is correct and useful for typing and state spectral analysis, but it cannot certify the parameterwise anti-constant hypothesis H5 from Section 7.

9.4 Separate Anti-Collapse Witness

Definition 9.20 (Affine Non-Constant Witness Space)**.** Let

$$\mathcal{H}_{ac} := L^2([0,1]), \qquad \mathcal{N} := \{c\mathbf{1} : c \in \mathbb{R}\}, \qquad h(s) := \sin(2\pi s).$$

Choose r with $0 < r < \|h\|_{L^2}$ and define the closed convex affine set

$$\mathcal{I}_{ac} := h + \{g \in \mathcal{H}_{ac} : \|g\|_{L^2} \le r\}.$$

Lemma 9.21 (The Anti-Collapse Witness Set Avoids Constants)**.**

$$\mathcal{I}_{ac} \cap \mathcal{N} = \varnothing.$$

Proof. The distance from h to the constant subspace $\mathcal{N}$ equals $\|h\|_{L^2}$ because h has mean zero and is orthogonal to constants. Since $r < \|h\|_{L^2}$, the radius-r ball around h does not intersect $\mathcal{N}$. □

Definition 9.22 (Anti-Collapse Witness Dynamics)**.** Let $\mathcal{U} = [0,1]$, choose $\beta \in (0, 1/2)$, and define

$$K_{ac} f := \beta f, \qquad \Gamma_{ac}(u) := \frac{u}{4} h.$$

Let P_{ac} denote the metric projection onto $\mathcal{I}_{ac}$, and define

$$T_u^{ac}(f) := P_{ac}(K_{ac} f + \Gamma_{ac}(u)).$$

Proposition 9.23 (Separate Witness for H5)**.** *The family $\{T_u^{ac}\}_{u \in \mathcal{U}}$ satisfies Hypotheses 3.1--7.3 of Section 1.*

Proof. 1. $\mathcal{H}_{ac}$ is a Hilbert space and $\mathcal{I}_{ac}$ is non-empty, closed, and convex.

2. $\|K_{ac}\| = \beta < 1$.

3. Completeness is inherited from $L^2([0,1])$.

4. $\mathcal{U} = [0,1]$ is compact and Γ_{ac} is continuous.

5. For every constant $c \in \mathcal{N}$ and every $u \in \mathcal{U}$, the value $T_u^{\mathrm{ac}}(c)$ belongs to $\mathcal{I}_{\mathrm{ac}}$, while $\mathcal{I}_{\mathrm{ac}} \cap \mathcal{N} = \varnothing$. Hence $T_u^{\mathrm{ac}}(c) \neq c$.

□

Remark 9.24. BaseCore therefore distinguishes two roles cleanly: the Golden Mean witness certifies typed operator correctness and state contraction; the affine anti-collapse witness certifies the stronger H5 burden needed for parameterwise non-collapse.

10 State Spectral Gap and Parameter-Family Persistence

Theorem 10.1 (State Spectral Gap)**.** *Fix $u \in \mathcal{U}$ in the typed computable model. Assume T_u is Fréchet differentiable at its fixed point f_u^*. Then*

$$\rho(DT_u(f_u^*)) \leq \|DT_u(f_u^*)\| \leq \left\|\widetilde{K}\right\| \leq \frac{1}{2}.$$

Consequently the state spectral gap satisfies

$$\Delta_{\mathrm{state}} \geq 1 - \left\|\widetilde{K}\right\| \geq \frac{1}{2}.$$

Proof. At every point where the projection P_E is Fréchet differentiable, its derivative has operator norm at most 1 because P_E is non-expansive. Since $\aleph_{\mathcal{B}} = j \circ P_E \circ A$,

$$DT_u(f_u^*) = D\aleph_{\mathcal{B}}\Big(\widetilde{K} f_u^* + \Gamma_{\mathcal{B}}(u)\Big) \circ \widetilde{K},$$

and therefore

$$\|DT_u(f_u^*)\| \leq \|D\aleph_{\mathcal{B}}\| \left\|\widetilde{K}\right\| \leq \frac{1}{2}.$$

The spectral-radius bound follows from $\rho(L) \leq \|L\|$ for bounded linear operators. □

Theorem 10.2 (Relaxation-Time Bound)**.** *Under the hypotheses of Theorem 10.1, any discrete-to-continuous relaxation convention of the form*

$$\tau_{\mathrm{state}} = -\frac{1}{\ln \rho}$$

with $\rho = \rho(DT_u(f_u^*))$ gives

$$\tau_{\mathrm{state}} \leq \frac{1}{\ln 2}.$$

Proof. Theorem 10.1 gives $\rho \leq 1/2$. Since $-\ln(\rho) \geq \ln 2$, the bound follows. □

Proposition 10.3 (Parameter-Family Persistence is not State Spectrum)**.** *Let $F : \mathcal{U} \to \mathcal{B}$ be the fixed-point family $F(u) = f_u^*$. Any persistence of parameter labels or family coordinates belongs to the variation of F over $\mathcal{U}$, not to the state-transition spectrum of $DT_u(f_u^*)$ for fixed u.*

Proof. The state-transition spectrum is defined for the linearization of T_u at fixed parameter u. Family-level transport instead concerns the map F between parameter space and fixed-point set. These are different operators on different domains. Therefore a family-persistence mode cannot be used to introduce a unit eigenvalue into the contraction spectrum of $DT_u(f_u^*)$. □

Remark 10.4 (What has been removed). The previous text incorrectly mixed strict contraction with a state-spectrum eigenvalue $\lambda = 1$. BaseCore no longer does that. Any neutral or persistent mode must now be interpreted as a parameter-family bookkeeping mode, not as an eigenmode of the contractive state Jacobian.

Part III
Identity, Rigidity, and Conditional Non-Simulability

11 Projective Systems and Inverse-Limit Identity

11.1 Projective Setup

Definition 11.1 (Projective System of State Spaces)**.** Let $\{S_t\}_{t\in\mathbb{N}}$ be a family of non-empty state spaces together with projection maps

$$\pi_{t+1\to t} : S_{t+1} \to S_t$$

satisfying the compatibility rule

$$\pi_{t+1\to t} \circ \pi_{t+2\to t+1} = \pi_{t+2\to t} \qquad \forall t \in \mathbb{N}.$$

Definition 11.2 (Identity as Inverse Limit)**.** The identity object associated with the projective system is

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t)_{t\in\mathbb{N}} \in \prod_{t\in\mathbb{N}} S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t \ \forall t \right\}.$$

Remark 11.3 (Identity neutrality). In BaseCore, $\mathcal{I}$ is a structural inverse-limit object only. It carries no automatic phenomenological, biological, or metaphysical interpretation.

Hypothesis 11.4 (Topological Regularity)**.** Each S_t is compact Hausdorff and each $\pi_{t+1\to t}$ is continuous and surjective.

Hypothesis 11.5 (NFD: Non-Finite Determination)**.** For every $t \in \mathbb{N}$, the canonical projection

$$p_t : \mathcal{I} \to S_t, \qquad p_t((x_k)_k) = x_t$$

is not injective. Equivalently,

$$\forall t \ \exists x, y \in \mathcal{I} \quad x \neq y \quad \text{and} \quad p_t(x) = p_t(y).$$

Remark 11.6 (Persistent future branching as an operational form of NFD). A sufficient operational strengthening of Hypothesis 11.5 is: for every t there exists $k > t$ and $x_t \in S_t$ such that the fiber of $\pi_{k\to t}$ over x_t contains at least two globally extendable compatible histories. BaseCore keeps the weaker NFD statement as the primary assumption because it is exactly what the non-locality theorem needs.

11.2 Non-Locality of Identity

Theorem 11.7 (Non-Locality of Identity under NFD)**.** *Assume Hypotheses 11.4 and 11.5. Then no canonical finite-time observation determines identity. More precisely, for every t the canonical projection*

$$p_t : \mathcal{I} \to S_t$$

is not injective, so identity is not recoverable from any finite slice through the projective observation maps.

Proof. This is exactly Hypothesis 11.5. The theorem restates the structural consequence in claim-boundary language: finite slices can participate in the inverse-limit object, but no single canonical slice determines it. □

Remark 11.8 (What is *not* proved here). BaseCore does not prove that no arbitrary set-theoretic injection $\mathcal{I} \to S_t$ can exist in every imaginable inverse system. It proves the narrower and correct claim that the canonical projective observations p_t are not injective under Hypothesis 11.5.

Proposition 11.9 (No Canonical Progressive Recovery under NFD)**.** *Assume Hypothesis 11.5. No finite canonical prefix map determines the inverse-limit identity object.*

Proof. If some finite canonical prefix determined identity, then in particular one of the canonical coordinate maps would become injective after finite truncation, contradicting Hypothesis 11.5. □

12 Rigidity under Admissible Perturbations

12.1 Metric Projective Assumptions

Assumption 12.1 (RIG-1: Common Metric Stages)**.** Both towers use the same nonempty complete metric stages (S_t, d_t) and continuous bonding maps. Use the weighted history metric below with $w_t \operatorname{diam}(S_t) \to 0$. This fixes the ambient space used for both Hausdorff directions.

Assumption 12.2 (RIG-2: Controlled Propagation)**.** For the fiber lifting constants $K_t > 0$ of RIG-3 define $q_t = w_{t+1}K_t/w_t$. Require $C = \sup_{0\le k\le t<\infty} \prod_{j=k}^{t} q_j < \infty$. In particular $q_t \le 1$ at each stage suffices.

Assumption 12.3 (RIG-3: Exact Fiber Lifts in Both Towers)**.** For each $f \in \{\pi_t, \widetilde{\pi}_t\}$, $u \in S_{t+1}$ and $v \in S_t$, there exists $u' \in S_{t+1}$ with $f(u') = v$ and $d_{t+1}(u, u') \le K_t d_t(f(u), v)$. The same K_t bounds both lifts. No continuous selection is required.

Assumption 12.4 (RIG-4: Typed Projection Defects)**.** Write $\pi_t = \pi_{t+1\to t}$ and $\widetilde{\pi}_t = \widetilde{\pi}_{t+1\to t}$. Require $\sup_{u\in S_{t+1}} d_t(\pi_t(u), \widetilde{\pi}_t(u)) \le \epsilon_t < \infty$.

Assumption 12.5 (RIG-5: Summable Deformation)**.** Require $E_w = \sum_{t\ge 0} w_t\epsilon_t < \infty$.

Definition 12.6 (Weighted Deformation Size)**.** The weighted deformation size is $\|\epsilon_\bullet\|_w = E_w$. When ϵ_t is the exact supremum projection defect this specifies a cost on the common-stage perturbations. An independently defined intervention cost needs its own quantitative relation to this sum.

Definition 12.7 (Weighted Product Metric)**.** For nonempty metric spaces (S_t, d_t) and weights $0 < w_t < \infty$, write

$$d_w(x, y) = \sup_{t\ge 0} w_t d_t(x_t, y_t), \qquad P_b = \{x \in \textstyle\prod_t S_t : d_w(x, b) < \infty\}.$$

The formula defines an extended metric on the product and an ordinary metric on each finite-distance component P_b.

Proposition 12.8 (Finiteness, Completeness and Compactness of Weighted Histories)**.** *The formula is finite on every pair in the full product exactly when the weighted diameters are bounded outside a finite set of stages. For finite-diameter stages this is equivalent to* $B = \sup_t w_t \operatorname{diam}(S_t) < \infty$. *For any nonempty inverse limit* I *and* $b \in I$*, it is finite on all of* I *exactly when* $I \subseteq P_b$*. If the coordinate spaces are complete and the bonding maps continuous,* P_b *and* $I \cap P_b$ *are complete. If* $w_t \operatorname{diam}(S_t) \to 0$*, the weighted metric induces the product topology and*

$$0 \le d_w(x, y) - \max_{0\le t\le N} w_t d_t(x_t, y_t) \le \sup_{t>N} w_t \operatorname{diam}(S_t).$$

In this case compact coordinate spaces give a compact inverse limit. Surjective bonding maps give a compatible history through each stage point.

Proof. The extended triangle inequality follows by taking suprema. A uniform bound outside finitely many stages suffices since each of the exceptional distances is individually finite. If no such bound exists, choose distinct stages t_j and pairs with weighted distances greater than j. Combining these pairs in two product points gives an infinite distance. The bound $B < \infty$ characterizes *boundedness* of the full-product metric. The inverse-limit condition follows from the triangle inequality through b.

A d_w-Cauchy sequence converges in each coordinate. Passing to each coordinate limit in the same uniform Cauchy bound proves convergence in d_w. The limit remains in P_b by the triangle inequality and in I by continuity of the bonding maps. With vanishing weighted diameter tails, a finite prefix controls the metric up to the displayed remainder, while each coordinate projection is $1/w_t$-Lipschitz. This proves the topology assertion. Conversely, on the full product with a finite metric, if the weighted diameters fail to vanish there are infinitely many coordinates with weighted diameter at least some $\eta > 0$. At any fixed product point, for each such coordinate one can change that coordinate by weighted distance greater than $\eta/3$, by the triangle inequality. The resulting sequence converges in the product topology but fails to converge in d_w. The converse may fail on a restricted set of compatible histories. The inverse limit is closed in the compact product because it is the intersection of the closed compatibility equations. Surjectivity gives successive lifts and the earlier coordinates are determined by projection. □

Example 12.9 (Two Distinct Tail Failures). On $S_t = [0, 1]$ with identity bonding maps and $w_t = 2^{-t}$, choose $d_t(s, u) = 4^t|s - u|$. The two constant histories 0 and 1 have distance $+\infty$. For a different failure, take $S_t = \{0, 1\}^t$ with prefix deletion and $d_t = 2^t$ times the discrete metric. With the same weights every distinct pair of histories has distance 1. The inverse limit has a compact product topology and a noncompact discrete metric topology. Flipping a bit at successively later levels gives product convergence to the unflipped history while all the weighted distances remain 1.

Theorem 12.10 (Conditional Stability of Inverse-Limit Identity)**.** *Under RIG-1 through RIG-5 the two limits $I = \mathcal{I}$ and $\widetilde{I} = \widetilde{\mathcal{I}}$ are nonempty. Every history in either limit has a compatible history in the other within distance CE_w. Thus $d_H(I, \widetilde{I}) \le CE_w$.*

Proof. Given a history x of π, set $y_0 = x_0$. Having chosen y_t, lift it under $\widetilde{\pi}_t$ at x_{t+1}. The exact lift y_{t+1} obeys

$$d_{t+1}(y_{t+1}, x_{t+1}) \le K_t(d_t(y_t, x_t) + \epsilon_t)$$

by the fiber estimate and $\pi_t(x_{t+1}) = x_t$. For $a_t = w_t d_t(y_t, x_t)$ this gives $a_{t+1} \le q_t a_t + q_t w_t \epsilon_t$, with $a_0 = 0$. Iteration bounds a_{t+1} by $\sum_{k=0}^{t}(\prod_{j=k}^{t} q_j) w_k \epsilon_k$, at most CE_w. The reverse construction uses the fiber estimate for π_t and gives the other directed Hausdorff bound. Each limit is nonempty by successive lifts from any S_0 point. Taking suprema proves the estimate in the common history metric. □

Corollary 12.11 (Conditional Isomorphism Requires Extra Structure)**.** *The Hausdorff bound alone is a statement about the distance between sets. A homeomorphism follows under the commuting-bijection hypotheses below.*

Proof. The following transport proposition gives the sufficient condition. Example 12.14 exhibits the failure of the unconditional implication. □

Proposition 12.12 (Transport by Commuting Coordinate Bijections)**.** *Suppose the weighted metric induces the product topology and homeomorphisms $h_t : S_t \to S_t$ satisfy $\widetilde{\pi}_t h_{t+1} = h_t \pi_t$. Then $H(x)_t = h_t(x_t)$ maps the limits homeomorphically, with inverse $H^{-1}(y)_t = h_t^{-1}(y_t)$. Its displacement in the common history space is at most $\sup_t w_t \sup_{z \in S_t} d_t(z, h_t(z))$.*

Proof. The displayed square and its inverse give exact compatibility of both images. Coordinate continuity in the product gives continuity of both induced maps. The displacement estimate follows coordinate by coordinate. □

Example 12.13 (A Ramified Tower with Positive Deformation and Coherent Transport). Take $S_t = (\mathbb{R}/\mathbb{Z}) \times \{0,1\}^t$ with the maximum of the circle distance (circumference one) and the discrete prefix distance. Let π_t keep the angle and delete the last bit, and take $w_t = 2^{-t}$. For an intervention at stage k with amplitude $0 < a < 1/2$, replace π_k by prefix deletion and angle shift $-a$. Set h_t to the identity for $t \le k$ and to angle shift a for $t > k$. The inverses shift by $-a$ on those same levels. The squares commute at k by cancellation and at every other level by commutation with prefix deletion. Each limit has at least two histories over every finite cut, since a later bit may be flipped.

All bonding maps are 1-Lipschitz and satisfy the fiber estimate with $K_t = 1$: set the required angle and prefix, retaining the last bit. Thus $q_t = 1/2$. The chosen dimensionless intervention cost is $E_w = w_k a$, and $\sup_x d_w(x, Hx) = w_{k+1} a = E_w/2$. More precisely, for $b = w_{k+1}$,

$$d_H(I, \widetilde{I}) = \frac{ba}{1+b} > 0.$$

Indeed an original history has constant angle θ and a perturbed one has angle φ up to k and $\varphi + a$ thereafter. If the two histories are at distance r, the circle triangle inequality gives $a \le r + r/b$ from stages 0 and $k+1$. Choosing identical bits and $\varphi = \theta - ba/(1+b)$ attains equality. That choice is reversible for every perturbed history and gives both directed distances. For the sector of these single jumps the infimum of the cost-to-deformation ratio is 2, since the ratio is $2(1+b)$ at stage k. This uses the declared projection-defect cost. An entropy-based cost requires a separate comparison.

Example 12.14 (Hausdorff Stability Without a Homeomorphism). Take $S_t = \{0, \dots, 2^t - 1\}$ with discrete distance scaled by $r > 0$, $w_t = 2^{-t}$, and $\pi_t(j) = \lfloor j/2 \rfloor$. For $t \ge 1$, change only the image of $j = 1$ to 1, keeping $\widetilde{\pi}_0 = \pi_0$. Both maps remain surjective and satisfy the fiber estimate with $K_t = 1$. The weighted intervention cost is $r \sum_{t \ge 1} 2^{-t} = r$. The all-zero history of the perturbed tower is isolated by the cylinder $y_1 = 0$ because each subsequent zero has exactly one lift. The original tower is the binary-prefix space and has no isolated histories. Thus the limits are not homeomorphic, even though the Hausdorff bound tends to zero with r. The perturbation hypotheses for the Hausdorff estimate all hold.

Definition 12.15 (Ontological Mass)**.** Let $E_w(\delta)$ denote the weighted energy of an admissible perturbation family δ. Define

$$M_\Omega := \inf \left\{ \frac{E_w(\delta)}{d_H(\mathcal{I}, \widetilde{\mathcal{I}}_\delta)} \;\middle|\; \delta \text{ admissible},\ d_H(\mathcal{I}, \widetilde{\mathcal{I}}_\delta) > 0 \right\},$$

with the convention $M_\Omega = +\infty$ if no admissible finite-energy perturbation yields positive deformation of the identity object.

Remark 12.16 (Interpretation discipline). M_Ω is a deformation-resistance scalar, not a metaphysical substance. BaseCore uses it only as a conditional invariant attached to admissible perturbation models.

13 Conditional Non-Simulability

Assumption 13.1 (NS-1: Target and Observable Family)**.** Fix a nonempty set of compatible histories $X \subseteq \mathcal{I}$, a family of queries J, and responses $F_j : X \to (Y_j, \rho_j)$. The query can specify an observable, a time horizon, or a declared intervention. All queries refer to the same

encoded history. The intervention protocol and parameters are known to both parties and fixed independently of the history, except for information explicitly included in the encoded state. Non-finite determination and deformation resistance are separate properties of a target, measured using their own definitions.

Assumption 13.2 (NS-2: Binary Representation Budget)**.** A deterministic encoder $c : X \to \{0,1\}^B$ has at most 2^B possible representations. The budget includes all history-dependent information accessible to the decoder. An adaptive binary transcript of depth at most B satisfies the same bound when its query tree is fixed in advance and the choices and termination are determined by its previous bits. An exact real parameter is outside this finite-alphabet class.

Assumption 13.3 (NS-3: Uniform Response Fidelity)**.** The decoders $g_j : \{0,1\}^B \to Y_j$ must satisfy $\rho_j(g_j(c(x)), F_j(x)) \leq \eta$ for every $x \in X$ and $j \in J$. The tolerance is in the declared response units, after normalizing them if necessary. Faithfulness of additional structures, such as the bonding maps or the intervention cost sector, imposes separate constraints.

Theorem 13.4 (Finite-Budget Obstruction by Response Separation)**.** *Under NS-1--NS-3, let $A \subseteq X$ be a finite set such that for each distinct $x, y \in A$ there is j with $\rho_j(F_j(x), F_j(y)) > 2\eta$. Then $|A| \leq 2^B$. For exact fidelity, a code exists exactly when the number of equivalence classes of $x \sim y$ defined by $F_j(x) = F_j(y)$ for all j is at most 2^B.*

Proof. If $c(x) = c(y)$, the decoded value for every given query is the same, so the triangle inequality gives $\rho_j(F_j(x), F_j(y)) \leq 2\eta$. Thus c is injective on A. At zero error, different response classes must use different representations. Conversely, assign a distinct representation to each class and decode its common response for each query. □

Theorem 13.5 (Approximation Barrier)**.** *On the binary prefix tower $S_t = \{0,1\}^t$ with weights $w_t = 2^{-t}$ and discrete stage metrics, use $Y = X = \{0,1\}^{\mathbb{N}}$ as the decoded history space, with $d_w(x, y) = 2^{-m}$ at the first differing bit $m \geq 1$. Among all encoders with at most 2^B states and decoders into X, the minimum worst-case d_w error is $2^{-(B+1)}$. Exact recovery of all finite prefix responses requires unbounded B as the query horizon increases.*

Proof. Keep the first B bits and pad the rest with zeros. The first possible mismatch is at $B+1$. For the lower bound, take all 2^{B+1} histories with arbitrary first $B+1$ bits and zero tail. Two share a code. An error strictly less than $2^{-(B+1)}$ would require the common decoded history to share the first $B+1$ bits with both, which is impossible. At horizon t all 2^t prefixes occur, so exact recovery of that response needs $B \geq t$, attained by storing those bits. □

Example 13.6 (One Exact Real Encodes Every Binary History). The map $b \mapsto \sum_{j \geq 1} 2b_j 3^{-j}$ is injective: at the first unequal bit j the leading difference is $2 \cdot 3^{-j}$ and the remaining tail is bounded by 3^{-j}. On the image, inverse decoding followed by each prefix projection recovers all the canonical observations from one exact real. As a separate intervention control, any declared family that leaves the target limit set fixed has $M_\Omega = +\infty$ by the defining convention, even if the allowed interventions have finite cost. Neither fact supplies a finite binary budget.

Remark 13.7 (What has been removed). The resource obstruction concerns the declared responses, metric tolerance and binary budget. Infinite-precision scalar encodings and finite-cost interventions require independent hypotheses before a binary-budget theorem can be applied to them. In the CCR program, non-finite determination and infinite resistance are separately specified; any stronger obstruction to faithful embeddings must state its category and preserved observations. Exact response classes determine whether a finite sufficient statistic exists for the declared queries.

14 Section Summary

1. Identity is an inverse-limit object, not a canonical finite slice.
2. Non-locality now depends explicitly on the NFD assumption.
3. Rigidity is stated as metric stability under explicit perturbation-model assumptions, not as automatic isomorphism.
4. Uniform observational fidelity obeys a binary-budget separation bound. The intervention-resistance class is determined separately.
5. BaseCore therefore preserves theorem strength where assumptions support it and weakens only those statements that were previously over-absolute.

Part IV
Regime Structure and Continuity Constraints

15 Formal Setup

15.1 Review of Identity Framework

We assume the identity framework of [4]. Let $(S_t, \pi_{t+1\to t})_{t\in\mathbb{N}}$ be a projective system of observable state spaces. The identity object is the inverse limit:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t)_t \in \prod_t S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t \;\; \forall t \right\}. \tag{1}$$

The ontological mass M_Ω quantifies resistance to identity deformation. We recall that:

- $M_\Omega = 0$: zero deformation-cost infimum;
- $0 < M_\Omega < \infty$: deformable persistence (finite rigidity);
- $M_\Omega = +\infty$: closed--rigid (CCR relative to the declared perturbation class).

15.2 Abstract Phenomenological Space

Definition 15.1 (Phenomenological Space)**.** A phenomenological space is a nonempty set $\mathcal{E}$ whose elements are called *phenomenological configurations*. No topology, metric, or algebraic structure is assumed a priori.

Remark 15.2. We do not claim that $\mathcal{E}$ represents ‘‘conscious states’’ or ‘‘qualia’’. It is an abstract mathematical space whose interpretation is not specified by the formalism. The space $\mathcal{E}$ is introduced purely as the codomain of a structural assignment; its ontological status is deliberately left open.

15.2.1 Why a Separate Space?

One might ask: why not use $\mathcal{I}$ itself as the phenomenological space? The answer is structural:

Proposition 15.3 (Endomorphisms Preserving All Canonical Projections)**.** *If $F : \mathcal{I} \to \mathcal{I}$ satisfies $p_t \circ F = p_t$ for every t, then $F = \mathrm{Id}$. This does not exclude endomorphisms with nonidentity coordinate actions.*

Proof. For each x, $F(x)$ and x have identical coordinates, hence are the same inverse-limit element. For contrast, on binary prefix spaces, bitwise complement defines an endomorphism induced by complement on every prefix; it commutes with the bonding maps but does not preserve each coordinate value. A separate codomain is a modeling choice, not a logical necessity for nonidentity maps. □

A separate codomain permits independent structural readouts; it is not forced by the universal property alone.

15.3 Phenomenological Assignment

Definition 15.4 (Phenomenological Assignment)**.** A phenomenological assignment is a family of total maps on the declared observable state spaces:

$$\phi_t : S_t \to \mathcal{E} \tag{2}$$

indexed by $t \in \mathbb{N}$.

Definition 15.5 (Induced Limit Assignment)**.** Given a phenomenological assignment $\{\phi_t\}$, the induced limit assignment, whenever it exists, is the map:

$$\Phi : \mathcal{I} \to \mathcal{E}, \quad \Phi((x_t)_t) := \lim_{t\to\infty} \phi_t(x_t), \tag{3}$$

The convergence statements below use the complete metric codomain of Hypothesis 16.3. Partial readouts may instead be studied on a specified common domain, with all projections and factorizations restricted to that domain.

15.3.1 Uniqueness of the Limit Assignment

Proposition 15.6 (Uniqueness and Uniform Error of the Limit Assignment)**.** *Under E1, each trajectory has a unique limit assignment and*

$$d_{\mathcal{E}}\big(\Phi(x), \phi_n(p_n x)\big) \le A_n \quad \textit{for all } x \in \mathcal{I}.$$

Proof. For $m > n$, the triangle inequality bounds the distance between the mth and nth readings by $\sum_{t=n}^{m-1} a_t$. Thus each sequence is Cauchy. Completeness supplies its limit, uniqueness follows from metric separation, and passage to the limit gives the bound. Since $A_n \to 0$ independently of x, convergence is uniform. □

15.4 Axioms for Phenomenological Coherence

We declare a controlled assignment model. E0 is a nontriviality convention, E1 supplies convergent readouts, and E2 is a separately checked non-factorization requirement.

Axiom 15.7 (E0: Non-Triviality)**.** The phenomenological space satisfies $|\mathcal{E}| \ge 2$, i.e., there exist at least two distinct phenomenological configurations.

Axiom 15.8 (E1: Controlled Asymptotic Compatibility)**.** In the complete metric codomain of Hypothesis 16.3, there are numbers $a_t \ge 0$, independent of the trajectory, with $\sum_{t=0}^{\infty} a_t < \infty$, such that

$$d_{\mathcal{E}}\big(\phi_{t+1}(p_{t+1}x), \phi_t(p_t x)\big) \le a_t \qquad (x \in \mathcal{I},\ t \ge 0).$$

Here $p_t : \mathcal{I} \to S_t$ is the canonical projection. Define the tail budget $A_n = \sum_{t=n}^{\infty} a_t$. Exact equality is the special case $a_t = 0$, not a requirement of E1.

Axiom 15.9 (E2: Non-Locality)**.** There exists no function $f : S_t \to \mathcal{E}$ (for any fixed t) such that for all trajectories $(x_t)_t \in \mathcal{I}$:

$$\Phi((x_t)_t) = f(x_t). \tag{4}$$

15.5 Status of the Assignment Conditions

The following results separate sufficient assumptions, a non-factorization requirement and their logical dependencies.

15.5.1 Nontriviality of the Codomain

Theorem 15.10 (E0 is Necessary)**.** *If $|\mathcal{E}| = 1$, then every phenomenological assignment is constant and no phenomenological classification is possible.*

Proof. If $\mathcal{E} = \{e_0\}$, any total assignment takes the value e_0 on every trajectory. It distinguishes no trajectories and factors through every canonical projection. It therefore cannot meet E2 on a nonempty domain. □

15.5.2 Controlled Convergence

Theorem 15.11 (Controlled Compatibility Is Sufficient)**.** *E1 guarantees existence of the limit with the error bound in Proposition 15.6. Absolute summability is sufficient, not necessary for an arbitrary convergent family.*

Proof. Existence and the bound were proved above. To disprove necessity in general, take readouts constant in the trajectory with values $s_n = \sum_{k=1}^{n} (-1)^{k+1}/k$ in $[0, 1]$. The even partial sums increase, the odd partial sums decrease, and their difference tends to zero, so they have a common limit. Their successive absolute increments form the divergent harmonic series; no summable uniform majorant exists. This example is a control for the necessity claim, not a realization of E2. □

15.5.3 Finite-Cut Non-Factorization

Theorem 15.12 (Non-Factorization Is an Independent Requirement)**.** *E2 excludes readouts determined by a single canonical finite cut. The inverse-limit construction and E1 alone do not imply E2.*

Proof. Take any nonempty inverse limit and a constant family $\phi_t = e_0$ in a codomain containing at least two points. E1 holds with zero tail, but $\Phi = e_0$ factors through every projection. Thus E2 must be checked separately. The nonempty construction in Theorem 15.15 checks it without assuming it. □

15.6 Logical Dependencies and Realizability

Theorem 15.13 (Logical Dependencies of the Assignment Conditions)**.** *On a nonempty trajectory domain, E2 implies E0. E1 does not imply E2. There exist assignments satisfying E0, the revised E1 and E2 simultaneously. Consequently these three conditions are not an independent minimal axiom basis as originally stated.*

Proof. If the codomain were a singleton, every assignment would be constant and factor through p_0, contradicting E2. The preceding constant-family control proves the second assertion. Theorem 15.15 proves the third by construction. E0 is retained as an explicit convention, although E2 entails it in the nonempty case. □

15.7 Rejected Alternatives

We briefly discuss axiom candidates that were considered and rejected.

15.7.1 Rejected: Continuity Axiom

One might require Φ to be continuous. This is not required by the chosen model because:

- The complete-metric model permits discontinuous finite readouts; continuity is a property of the regular-readout subclass, not part of the unrestricted assignment condition;
- Continuity is derived from controlled approximation by continuous finite readouts; a scalar CCR label alone is insufficient.

15.7.2 Rejected: Surjectivity Axiom

One might require Φ to be surjective. This is not required by the chosen model because:

- A readout need not exhaust a codomain chosen to include other regimes;
- Surjectivity is a strong structural claim not derivable from identity constraints.

15.7.3 Rejected: Injectivity Axiom

One might require Φ to be injective. This is not required by the chosen model because:

- Many trajectories may share phenomenological configurations;
- Injectivity is not imposed on the full assignment class; it is a consequence in the concrete coercive realization. An injective map alone is not necessarily a topological embedding.

15.8 Structural Interpretation

Axiom	**Structural Meaning**	**Status**
E0	Nontrivial codomain	E2 consequence
E1	Summable increments	Uniform limit
E2	Finite-cut non-factorization	Required

15.9 Closure of Section 2

The limit is uniquely defined with a uniform tail bound. E2 remains a substantive non-factorization requirement; E0 is redundant when E2 holds on a nonempty domain. Exact compatibility cannot coexist with E2, whereas the controlled condition has the explicit realization below. No existence or minimality claim is obtained merely by naming an axiom.

15.10 Phenomenological Neutrality Clause

Throughout this paper, the phenomenological space $\mathcal{E}$ is treated as a *purely abstract topological and metric structure*. No semantic, psychological, biological, or experiential interpretation is assumed or required for any of the formal results derived herein.

The term ‘‘phenomenological’’ refers exclusively to the role of $\mathcal{E}$ as an abstract codomain of structural compatibility for inverse-limit identities, and does not presuppose the existence of consciousness, subjective experience, or any particular phenomenological content.

All theorems, constructions, and stability results presented in this work are therefore statements about abstract mathematical objects only. Any interpretation of $\mathcal{E}$ in relation to experiential, cognitive, or phenomenological phenomena lies strictly outside the formal scope of this paper.

15.11 Exact Compatibility Obstruction

Proposition 15.14 (Exact Compatibility Forces Finite-Cut Factorization)**.** *On a nonempty trajectory domain, if $\phi_{t+1}(p_{t+1}x) = \phi_t(p_t x)$ for every x, t, then the limit assignment satisfies $\Phi = \phi_0 \circ p_0$. It cannot satisfy E2. The controlled E1 above replaces that exact equality, rather than assuming it together with E2.*

Proof. Induction makes all readings equal to the first. Their limit is therefore $\phi_0(p_0 x)$. For partial readouts with a declared common domain, the same argument holds on that domain. If a total map on S_0 is required, extend the defined first-cut values arbitrarily outside the projected common domain, using any fixed element of the nonempty codomain. Thus partiality alone does not remove the obstruction. □

15.12 A Nonempty Controlled Assignment Model

Theorem 15.15 (Ternary Realization of E0--E2)**.** *Let $S_t = \{0,1\}^{t+1}$ with bonding maps that delete the last bit. Identify its inverse limit with $\mathcal{I} = \{0,1\}^{\mathbb{N}}$ and let $\mathcal{E} = [0,1]$. Define*

$$\phi_t(b_0, \ldots, b_t) = \sum_{j=0}^{t} \frac{2b_j}{3^{j+1}}, \qquad \Phi(b) = \sum_{j=0}^{\infty} \frac{2b_j}{3^{j+1}}.$$

These assignments satisfy E0, controlled E1, and E2, with

$$a_t = \frac{2}{3^{t+2}}, \qquad A_t = \frac{1}{3^{t+1}}, \qquad \sup_b |\Phi(b) - \phi_t(p_t b)| = A_t.$$

For distinct histories first differing at position m, put $d_{\mathcal{I}}(b,c) = 3^{-(m+1)}$, and put $d_{\mathcal{I}}(b,b) = 0$. Then

$$d_{\mathcal{I}}(b,c) \le |\Phi(b) - \Phi(c)| \le 3 d_{\mathcal{I}}(b,c).$$

The domain is compact and complete, and the readout is injective and bi-Lipschitz onto its image. Thus a global lower bound is realized here rather than inserted without a witness.

Proof. Every infinite bit sequence defines a compatible prefix history, so the domain is nonempty. Geometric summation puts every value in $[0,1]$ and gives the increment and tail bounds. The all-one tail attains the tail bound. For any fixed t, a zero history and a history with its sole one at position $t+1$ have the same prefix but images differing by $2/3^{t+2} > 0$. No function of p_t determines Φ, proving E2 for every finite cut.

At the first differing position m, the principal difference has magnitude $2/3^{m+1}$; the remaining tail has magnitude at most $1/3^{m+1}$. Subtraction and addition give the two bounds. This metric is the weighted supremum of the discrete prefix metrics with weights $3^{-(t+1)}$; bounded prefix diameters ensure finiteness. A Cauchy sequence for $d_{\mathcal{I}}$ eventually stabilizes each finite prefix, which defines its limit. Prefix cylinders give finite covers of arbitrarily small diameter; the space is totally bounded and complete, hence compact. The two bounds establish injectivity and quantitative continuity. Compactness and continuity give properness into the Hausdorff codomain, and a later-bit flip in every neighborhood gives local non-collapse. Thus this realization also supplies all four clauses of the stated readout regularity hypothesis. A scalar real-valued readout still encodes the complete sequence: E2 excludes the specified canonical finite cuts, not every possible coding or every finite-dimensional representation. □

Theorem 15.16 (Sharp Finite-Cut Approximation Error)**.** *For the ternary realization and an arbitrary decoder $g : S_t \to \mathbb{R}$, let*

$$R_t(g) = \sup_{b \in \mathcal{I}} |\Phi(b) - g(p_t b)|.$$

Then the exact minimax value is

$$\inf_g R_t(g) = \frac{1}{2\,3^{t+1}}.$$

It is attained by $g_*(s) = \phi_t(s) + 1/(2\,3^{t+1})$.

Proof. For a fixed prefix s, the all-zero and all-one suffixes give endpoints $\phi_t(s)$ and $\phi_t(s) + A_t$. For any number $g(s)$, the triangle inequality forces at least one endpoint error to be at least $A_t/2$. Conversely all readout values in that fiber lie between those endpoints, so their midpoint has error at most $A_t/2$. Both endpoints belong to the domain, establishing the exact optimum and attainment. The bound is positive for each fixed finite budget and tends to zero as t increases; it gives no positive uniform error floor across increasing resolutions. □

Proposition 15.17 (Sharp Error for Arbitrary Binary Codes)**.** *For the ternary assignment above, let an encoder choose at most* 2^B *states as an arbitrary function of the whole history, and let each state decode to any real number. The minimum worst-case absolute error is* $1/(2 \cdot 3^B)$ *for every integer* $B \geq 0$.

Proof. The cylinders of depth B have intervals of length 3^{-B} between their two attainable endpoints. Different intervals have gaps at least 3^{-B}: after the first differing bit $j \leq B$, the maximum opposing tail is 3^{-j}. Thus all 2^{B+1} endpoints, including the depth-zero endpoints 0 and 1, are pairwise separated by at least 3^{-B}. Two share a code and the triangle inequality forces error at least $3^{-B}/2$. Encoding the B-bit prefix and decoding to its interval midpoint attains the bound. □

15.13 Common Readout on Two Coherently Transported Towers

Theorem 15.18 (Joint Realization of Transport and Controlled Readouts)**.** *Use circle-prefix stages* $S_t = \mathbb{T} \times \{0,1\}^t$, *stage metric* $\max(\rho, \mathbf{1}_{\text{different prefix}})$, *and weights* $w_t = 3^{-t}$, *where* ρ *is circle distance with circumference one. Take the usual truncation tower* I_0 *and its variant* I_1 *with angle shift* $-a$ *at the single bonding level* k, *for* $0 < a < 1/2$. *Write* $\beta = 3^{-(k+1)}$. *In the common weighted history space put* $Z = I_0 \cup I_1$. *Let the complete codomain be* $\mathcal{E} = \mathbb{T}^2 \times [0,1]$ *with the maximum of the first circle distance,* β *times the second circle distance, and the absolute scalar difference. Define*

$$F(x) = \left(\theta_0(x), \theta_{k+1}(x), \sum_{j \geq 1} 2b_j(x) 3^{-j} \right).$$

Then $d_w(x,y) \leq d_{\mathcal{E}}(F(x),F(y)) \leq 3d_w(x,y)$ *on all of* Z. *On each tower the restriction satisfies E0--E2 with* $a_t = 2 \cdot 3^{-(t+1)}$ *and uniform tail* $A_t = 3^{-t}$. *The coordinate homeomorphism* $H : I_0 \to I_1$ *shifts angles at stages* $t > k$ *by* a. *With projection-defect cost* $E_w = 3^{-k}a$, *one has*

$$d_w(x, Hx) = d_{\mathcal{E}}(F(x), F(Hx)) = \beta a, \qquad d_H(I_0, I_1) = \frac{\beta a}{1+\beta}.$$

Proof. In each tower the angle is constant on the two blocks $t \leq k$ and $t > k$. Thus the angular part of the history distance is exactly the maximum of the two weighted angular distances in the codomain. The binary part is 3^{-j} at first differing bit j. The leading ternary term and its tail give lower and upper constants 1 and 3 for that part. Each tower is closed in the compact metric product, so their union is complete.

For a declared tower $s \in \{0,1\}$ the finite observation $(\theta_t, b_1, \ldots, b_t)$ gives the initial angle $\theta = \theta_t - sa\mathbf{1}_{t>k}$. The stage readout is $(\theta, \theta + sa, \sum_{j=1}^t 2b_j 3^{-j})$. The phase rule uses the declared bonding family, and on both towers the successive readout increment is bounded by a_t, with exactly the stated uniform tail. A later-bit flip proves E2, since both phase coordinates stay

fixed while the scalar changes. The codomain is nontrivial, proving E0. Within either fixed-angle fiber the sharp binary-code result applies to the scalar readout.

The coordinate squares commute because the sole change of phase at stage $k+1$ cancels the shift $-a$ in its bonding map. The coordinatewise inverse shifts by $-a$ on the later block. For fixed bits, F measures the shift βa exactly. In the comparison of sets, a pair at distance r has $a \le r + r/\beta$ by the triangle inequality at stages 0 and $k+1$. Equal bit sequences and an initial phase offset $-\beta a/(1+\beta)$ for the second history attain this bound in both directions. □

15.14 Controls Separating Convergence, Continuity, and Invariance

Proposition 15.19 (Setwise Invariance Does Not Supply Continuity)**.** *Controlled E1 and E2 can hold for a discontinuous assignment, even if the identity domain is fixed under every declared intervention.*

Proof. Take $S_t = [0,1] \times \{0,1\}^{t+1}$, with prefixes on the bit coordinates, and readouts

$$g_t(s,b) = \tfrac{1}{2}\mathbf{1}_{\{s>0\}} + \sum_{j=0}^{t} \frac{b_j}{3^{j+1}}.$$

The increments are bounded by $3^{-(t+2)}$, so they satisfy controlled E1 in $[0,1]$. Later bits prove E2 as above. With all bits zero, values at $s=0$ and $s=1/n$ differ by $1/2$ although their product-metric distance tends to zero. The limit is discontinuous. Let the declared intervention family consist only of the identity: the domain is setwise fixed, which does not alter the counterexample. This is a control for the asserted inference from invariance to continuity; no claim of full CCR or operational-criterion membership is made for it. □

Proposition 15.20 (Setwise and Pointwise Transport Are Distinct)**.** *Preserving an identity domain as a set does not guarantee equal values of a common readout on transported states.*

Proof. On $Z = \{0,1\}$, let $h(0) = 1$, $h(1) = 0$ and $F(x) = x$. The transport preserves Z exactly but changes both readout values. Thus setwise rigidity cannot replace the pointwise or readout-natural comparison hypotheses used below. □

16 Compatibility with Identity Regimes

16.1 Induced Constraints from Identity Rigidity

Let $(\mathcal{I}, M_\Omega)$ be an identity object with ontological mass M_Ω. We now study how M_Ω constrains the set of admissible phenomenological assignments.

Definition 16.1 (Phenomenological Compatibility Class)**.** For a declared identity domain and codomain, $\mathcal{P}(M_\Omega)$ consists of maps $\Phi : \mathcal{I} \to \mathcal{E}$ that are limits of some family $\{\phi_t\}$ satisfying E1, with E0 and E2 also satisfied. The readout family and its tail control are part of the realization data. The notation retains the mass index for comparisons, but no mass-dependent restriction follows from E0--E2 alone.

Remark 16.2. The compatibility class records choices of domain, codomain and readout family. A numerical mass label does not by itself determine those choices or their transport under perturbations.

16.2 Metric Structure for Quantitative Comparisons

Quantitative comparisons use a declared metric. The following statements identify the role and limits of that choice.

Hypothesis 16.3 (H3: Metric on $\mathcal{E}$)**.** The phenomenological space $\mathcal{E}$ admits a metric $d_{\mathcal{E}}$ such that $(\mathcal{E}, d_{\mathcal{E}})$ is a complete metric space.

16.2.1 Role of the Metric

Theorem 16.4 (Metric Data Required by Quantitative Formulas)**.** *The distance-based fragmentation and loss functionals used here require a declared metric, or a pseudometric with a stated quotient convention. Topological continuity can be defined without a metric. Completeness is used for the Cauchy-limit guarantee of E1.*

Proof. Each quantitative functional explicitly contains $d_{\mathcal{E}}$, so it is undefined until that datum is supplied. Continuity can alternatively be defined through inverse images of open sets. A pseudometric can identify zero-distance states by a quotient, but that changes the domain of class labels and must be specified. These facts justify the chosen quantitative model without claiming that all alternatives are impossible. □

16.2.2 Completeness as a Limit Guarantee

Proposition 16.5 (Completeness and Existence of Limits)**.** *Without completeness, the conclusion that every E1-type Cauchy family has its limit in the declared codomain can fail. Completeness is not necessary for a family whose limit is already known to belong to that codomain.*

Proof. In $\mathbb{Q} \cap [0, 2]$ with its usual metric, use dyadic rational approximations $q_n = 2^{-n}\lfloor 2^n\sqrt{2}\rfloor$ as constant-in-trajectory readings. The increments are bounded by 2^{-n}, a summable sequence, but their real limit $\sqrt{2}$ is outside the codomain. This isolates the role of completeness when supplying a general existence guarantee; it makes no assertion that a convergent constant family needs completeness. □

16.2.3 Alternatives Considered and Rejected

Alternative 1: Topology without Metric. One might use a general topological space. This is not adopted in the present quantitative model because:

- Fragmentation requires quantitative distance, not just open sets;
- Loss capacity requires a sup over distances;
- The downstream instability analysis completed in *Null-Regime Instability* requires metric quotients.

Alternative 2: Pseudometric. One might allow $d_{\mathcal{E}}(e_1, e_2) = 0$ for $e_1 \neq e_2$. This is not adopted in the present quantitative model because:

- It would conflate distinct phenomenological configurations;
- Cardinality and metric separation are different requirements; zero-distance classes would require a declared quotient before distinguishability is counted.

Alternative 3: Non-Complete Metric. This is rejected by Proposition 16.5.

16.3 Phenomenological Continuity

Definition 16.6 (Phenomenological Continuity)**.** A phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$ is *structurally continuous* if for every sequence $(x^{(n)})_n$ in $\mathcal{I}$ converging in the d_w metric:

$$d_w(x^{(n)}, x) \to 0 \implies d_{\mathcal{E}}(\Phi(x^{(n)}), \Phi(x)) \to 0. \tag{5}$$

Remark 16.7. Structural continuity is derived under the finite-readout conditions of Theorem 17.7. It is distinct from invariance under an external intervention.

16.4 Phenomenological Fragmentation

Definition 16.8 (Phenomenological Fragmentation)**.** A phenomenological assignment Φ is *fragmentable* if there exists a perturbation $\{\epsilon_t\}$ with finite weighted energy such that:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0 \tag{6}$$

for corresponding identity trajectories $x \in \mathcal{I}$ and $\tilde{x} \in \tilde{\mathcal{I}}$.

Remark 16.9. Fragmentation captures the vulnerability of phenomenological structure to perturbations. It is the phenomenological analog of identity deformability.

16.5 Relationship Between Metrics

Proposition 16.10 (Continuity from Controlled Finite Readouts)**.** *Write $g_n = \phi_n \circ p_n$. Under E1,*

$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \le 2A_n + d_{\mathcal{E}}(g_n(x), g_n(y)).$$

If every g_n is continuous, then Φ is continuous. If every g_n is L-Lipschitz with a common L, then Φ is L-Lipschitz.

Proof. Insert $g_n(x)$ and $g_n(y)$ and apply the uniform error bound. For continuity, first make $2A_n$ smaller than half the prescribed error, then use continuity of that fixed g_n. For the Lipschitz conclusion use the same bound for all n and let $n \to \infty$. □

16.6 Closure of Section 3

The quantitative formulas use a declared complete metric codomain. Continuous finite readouts and controlled tails supply a continuity theorem. Cross-system comparisons additionally require a declared common domain and transport; neither mass labels nor distances between sets substitute for those data.

17 Impossibility Theorems

17.1 Φ-Regularity Hypothesis

Before proving the main impossibility results, we isolate the global lower-bound regularity condition used by the fragmentation argument. The condition is not derived from properness, local Lipschitz regularity, and non-collapse alone; it is carried as an explicit hypothesis.

Hypothesis 17.1 (Φ-Regularity Lower Bound)**.** Let $(\mathcal{I}, d_w)$ be the identitary metric space from *Rigid Identity* and let $(\mathcal{E}, d_{\mathcal{E}})$ be the abstract phenomenological metric space. A phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$ is said to satisfy Φ-regularity when it satisfies:

1. (Properness) For every compact $K \subset \mathcal{E}$, $\Phi^{-1}(K)$ is compact in $\mathcal{I}$.

2. (Local Lipschitz regularity) For every $x \in \mathcal{I}$, there exist $\varepsilon > 0$ and $L_x > 0$ such that for all $y, z \in B_\varepsilon(x)$:
$$d_{\mathcal{E}}(\Phi(y), \Phi(z)) \le L_x \cdot d_w(y, z). \tag{7}$$

3. (Non-collapse) For every $x \in \mathcal{I}$ and every $r > 0$, the restriction of Φ to $B_r(x)$ is not constant.

4. (Global lower Lipschitz bound) There exists a constant $C > 0$ such that for all $x, y \in \mathcal{I}$,
$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \ge C \cdot d_w(x, y). \tag{8}$$

Caveat 17.2 (Status of Φ-regularity). The fourth clause is an additional structural assumption, not a consequence of the first three clauses. The counterexample in Appendix A.5 shows that properness, local Lipschitz regularity, and non-collapse do not imply any uniform lower Lipschitz constant.

Hypothesis 17.3 (Audited Perturbation Comparisons)**.** Declare a finite-cost intervention class $\mathcal{V}$, a common metric space (Z, d_Z) containing the unperturbed and perturbed identity domains, and transports $h_\varepsilon : \mathcal{I} \to \mathcal{I}_\varepsilon \subseteq Z$. Readouts are restrictions of one declared map $F : Z \to \mathcal{E}$: $\Phi = F|_{\mathcal{I}}$ and $\Phi_\varepsilon = F|_{\mathcal{I}_\varepsilon}$. The following branches are alternatives, used separately. The positive-displacement branch has some $\varepsilon \in \mathcal{V}$ and $x \in \mathcal{I}$ with $d_Z(x, h_\varepsilon x) > 0$. The zero-displacement branch has $h_\varepsilon x = x$ for every $\varepsilon \in \mathcal{V}$ and $x \in \mathcal{I}$. Any use of the lower Lipschitz bound in a comparison requires that bound for F on the common comparison domain, not merely within one identity object.

Remark 17.4 (Burden linking rigidity and readout comparisons). The scalar M_Ω does not itself supply the transports or a common readout. Relating its finite and infinite regimes to these two branches is a separate model obligation. Setwise invariance does not imply pointwise fixed transport, and a lower bound within one domain does not compare unrelated assignments on different domains.

17.2 Fragmentation Theorem

Theorem 17.5 (Fragmentation under an Audited Displacement)**.** *Let an assignment satisfy E0--E2. Assume the positive-displacement branch of Hypothesis 17.3 and a lower Lipschitz constant $C > 0$ for F on its common comparison domain. Then, for the witnessing pair,*

$$d_{\mathcal{E}}(\Phi(x), \Phi_\varepsilon(h_\varepsilon x)) \geq C d_Z(x, h_\varepsilon x) > 0.$$

Thus the assignment is fragmentable. This applies to a finite-M_Ω model when that model supplies the stated comparison data.

Proof. Both arguments lie in the declared domain of F. Its lower bound gives the inequality, and the intervention has finite cost by membership in $\mathcal{V}$. E2 is not used to infer that every deformation changes a readout: the lower bound and the positive witness perform that work. □

Corollary 17.6 (Fragmentability and Continuity Are Distinct)**.** *The positive-displacement conclusion does not establish a topological discontinuity. A continuous assignment may be fragmentable under a state-changing intervention.*

Proof. The ternary realization has a continuous assignment. Declare a finite-cost intervention that flips one bit while retaining the common readout. Its value changes by a positive amount, so the assignment is fragmentable in the stated sense while remaining continuous. This does not assert a full CCR realization. □

17.3 Forced Continuity Theorem

Theorem 17.7 (Continuity and Conditional Perturbation Invariance)**.** *Under E1, if each finite readout $g_n = \phi_n \circ p_n$ is continuous, then Φ is continuous, irrespective of M_Ω. Under the zero-displacement branch of Hypothesis 17.3,*

$$\Phi_\varepsilon(h_\varepsilon x) = \Phi(x) \qquad (\varepsilon \in \mathcal{V},\ x \in \mathcal{I}).$$

For a CCR model the invariance conclusion therefore holds when its finite-cost interventions admit that branch. CCR alone is not a proof of continuity or of readout naturality.

Proof. Continuity follows from Proposition 16.10. For invariance use the common readout and the fixed transport: $\Phi_\varepsilon(h_\varepsilon x) = F(h_\varepsilon x) = F(x) = \Phi(x)$. Neither argument identifies invariance under external perturbation with continuity along convergent states. □

Corollary 17.8 (Invariance on the Audited Zero-Displacement Branch)**.** *A CCR model satisfying the zero-displacement comparison hypothesis has invariant assignment values under all interventions in its declared finite-cost class.*

Proof. Apply the common-readout equality in Theorem 17.7 for each declared intervention. □

17.4 Dichotomy Theorem

Theorem 17.9 (Conditional Classification by Audited Branches)**.** *Suppose a model supplies the positive-displacement branch and lower bound whenever $M_\Omega < \infty$, and supplies continuous finite readouts and the zero-displacement branch whenever $M_\Omega = +\infty$. Then its finite regime is fragmentable and its infinite regime has a continuous, perturbation-invariant assignment. The mass cases are exclusive; fragmentability and topological continuity need not be exclusive properties.*

Proof. Apply Theorems 17.5 and 17.7 with their respective comparison hypotheses. Exclusivity of the cases follows from the numerical alternatives for M_Ω, not from a false identification of a changed readout with a discontinuity. Establishing the branch hypotheses for a given rigidity model remains an explicit obligation. □

18 Classification of Phenomenological Regimes

18.1 Conditional Classification

The results give a conditional branch classification after the comparison and readout hypotheses have been checked:

Identity Class	M_Ω	**Admissible Phenomenology**
Null label	$= 0$	Domain/readout link required
Deformable	$(0, \infty)$	Positive witness and lower bound
Rigid (CCR)	$= +\infty$	Regular readout / fixed transport

18.2 Structural Implications

Proposition 18.1 (The Mass Label Does Not Define Assignment Existence)**.** *For fixed identity domain, codomain and readout family, the class defined by E0--E2 is independent of a numerical label M_Ω. A claim that $M_\Omega = 0$ excludes assignments needs an additional link to the domain or admissible readouts.*

Proof. The displayed definition of $\mathcal{P}(M_\Omega)$ tests E0--E2, none of which contains that scalar. For an empty trajectory domain E2 fails, since a function on a finite cut would agree vacuously on every trajectory. Deriving such domain loss from $M_\Omega = 0$ is not one of E0--E2 and cannot be inserted without a model. □

Proposition 18.2 (Deformable Branch with Comparison Data)**.** *For $0 < M_\Omega < \infty$, the positive-displacement and lower-bound hypotheses of Theorem 17.5 imply fragmentability.*

Proof. Apply the quantitative inequality of Theorem 17.5 to its witnessing finite-cost intervention. □

Proposition 18.3 (Rigid Branch with Comparison Data)**.** *For $M_\Omega = +\infty$, continuous finite readouts under E1 imply continuity, and the zero-displacement branch implies perturbation invariance. These are the respective hypotheses of Theorem 17.7.*

Proof. Apply the two conclusions of Theorem 17.7 with their separate hypotheses. □

19 Extended Proofs

19.1 Proof of Fragmentation Theorem (Extended)

Full Proof of Theorem 17.5. Use the specified common space Z, map F and finite-cost transport h_ε. Let x be its positive-displacement witness and put $\delta = d_Z(x, h_\varepsilon x) > 0$. Apply the lower bound on that same common domain to obtain readout displacement at least $C\delta$. No comparison of set distances is substituted for this pointwise distance. If both readouts are estimated with errors at most η, the triangle inequality gives estimated separation at least $C\delta - 2\eta$; it remains positive when $C\delta > 2\eta$. This is a bound on pair separation, not a statement that a particular endpoint lies outside a null ball. □

19.2 Proof of Forced Continuity Theorem (Extended)

Full Proof of Theorem 17.7. For a state x and error $\epsilon > 0$, choose n with $2A_n < \epsilon/2$. Continuity of g_n gives a neighborhood of x where $d_{\mathcal{E}}(g_n(x), g_n(y)) < \epsilon/2$. The two tail errors and the finite-readout difference bound the limiting difference by ϵ. Independently, a zero-displacement comparison satisfies $F(h_\varepsilon x) = F(x)$ exactly. Setwise invariance of the identity object alone would not imply either of these steps. □

19.3 Structural Necessity Lemma

Lemma 19.1 (Inheritance under Audited Fixed Transport)**.** *An assignment satisfying the zero-displacement comparison branch inherits invariance under that declared intervention class. This applies to CCR realizations only after checking that branch.*

Proof. The comparison readout is the restriction of F, and every specified transport fixes its argument. Therefore its value is unchanged. This proves the restricted inheritance statement without assuming that scalar rigidity determines arbitrary assignments. □

Part V
Null-Regime Structural Exclusion

20 Compatibility, Witnesses, and Non-Nullity

20.1 Compatibility Operator

Definition 20.1 (Compatibility Operator)**.** Declare a complete metric comparison domain (Z, d_Z), a complete metric poset $(\mathcal{E}, \preceq, d_{\mathcal{E}})$ with bottom $\varnothing_\phi$, and maps $\Phi_S : D_S \to \mathcal{E}$. The audited readout $\Phi : Z \to \mathcal{E}$ is defined on its declared pair class $W \subseteq Z \times Z$. Extension and isomorphism transports $j : D_S \to D_{S'}$ satisfy $\Phi_{S'} \circ j = \Phi_S$ when claimed. On W, a certified $C > 0$ satisfies

$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq C d_Z(x, y).$$

One may choose $Z = \mathcal{I}$ for history comparisons. Comparing whole identity objects requires a separate declared embedding/metric; $\Phi(\mathcal{I})$ is not a point value when the domain is the set $\mathcal{I}$.

A separated extension witness supplies two distinct defined points in a common comparison domain, not merely a relabeled copy under exact transport. CCR does not supply such a pair or lower constant.

Definition 20.2 (Phenomenological Attractor Set)**.** Use $\mathrm{Attr}(S)$ for structurally stable readout values under a declared perturbation/extension family, as defined below. Temporal attraction would additionally require a dynamics and a basin; neither is implied by this notation.

Definition 20.3 (Phenomenological Stability)**.** A value $e = \Phi_S(z)$ is structurally stable if a finite modulus $\omega(\rho) \to 0$ as $\rho \downarrow 0$ controls $d_{\mathcal{E}}(\Phi_{S'}(Pz), e) \le \omega(\rho)$ for all declared perturbation transports P of cost at most ρ. Exact persistence instead requires equality. Stability does not force equality at a nonzero perturbation budget.

Lemma 20.4 (Bridge Lemma)**.** *For any defined $e \in \mathcal{E}$, distance $d_{\mathcal{E}}(e, \varnothing_\phi) = 0$ implies $e = \varnothing_\phi$. No CCR assumption is needed.*

Proof. This is metric separation. □

20.2 Witness-Relative Null Separation

Theorem 20.5 (Conditional Null Separation)**.** *Suppose $(z_0, z_1) \in W$, $d_Z(z_0, z_1) = \epsilon > 0$, and the lower constant in Definition 20.1 is $C > 0$. Their images cannot both equal null, and*

$$\max_{i=0,1} d_{\mathcal{E}}(\Phi(z_i), \varnothing_\phi) \ge C\epsilon/2.$$

The historical instability label denotes joint exact null exclusion here, not dynamical instability of an individual null endpoint.

Proof. Joint nullity would give mutual image distance zero, contradicting $C\epsilon > 0$. The triangle inequality bounds that mutual distance by the sum of their distances to null, at most twice their maximum. □

Corollary 20.6 (Conditional Forced Non-Nullity)**.** *Under the separated-pair hypotheses, at least one witness image is non-null. If one endpoint is fixed at null, the other has null distance at least $C\epsilon$. CCR alone does not select a non-null endpoint or guarantee a stable regime.*

Proof. The first assertion follows from joint null exclusion. In the anchored case, mutual image distance is the other endpoint's distance to null, so the lower bound applies directly. □

Example 20.7 (External Null Witness in a Common Readout Domain). Fix $k \ge 0$ and $0 < a < 1/2$. Use the weighted circle-prefix history metric with weights 3^{-t}, the undeformed tower I_0, the single-jump tower I_1 (bonding shift $-a$ at level k), and $Z = I_0 \cup I_1$. Set $\beta = 3^{-(k+1)}$. The common readout F takes the initial angle, the angle at $k+1$, and the ternary code of the bits, with codomain metric $\max(\rho_0, \beta\rho_1, |\Delta\mathrm{code}|)$. The two angular blocks give the first two terms of d_w, while the ternary leading term minus its tail bounds the binary part from below. Thus $C = 1$ holds on the entire compact union. Order the codomain by equality together with $e_0 = (0, 0, 0)$ below every element.

Declare internal interventions to be simultaneous rotations of all stages of I_0, with cost the absolute shortest angular shift. Every such map has its inverse and sends I_0 onto itself. This gives infinite resistance in this sector by the Hausdorff convention, despite finite-cost nonidentity transports. The pointwise readout changes under a nonzero rotation. A sector admitting the single-jump deformation instead has a finite resistance witness with projection-defect cost $3^{-k}a$. The two sectors specify different admissible operations.

Declare as a separate extension the addition of the single-jump branch to I_0. It is the union of two compact projective limits; it is also realized by the disjoint union of the two stage towers

with branch-preserving projections. Embed its histories in Z by forgetting the tag, which is injective at the limit since the phases across k distinguish the two branches. The inclusion $j : I_0 \hookrightarrow Z$ preserves the restricted readout exactly. Take $z_0 = j(x)$ with zero phase and all-zero bits and z_1 on the added branch with the same initial phase and bits. Then

$$d_Z(z_0, z_1) = \beta a, \qquad F(z_0) = e_0, \qquad d_{\mathcal{E}}(F(z_1), e_0) = \beta a.$$

The extension is positive in Hausdorff distance, $\beta a/(1+\beta)$, using the two angular blocks. The transported point and the added endpoint are distinct. The scalar ternary readout alone, which discards both angles, assigns them the same value and fails the positive lower bound on this pair. The example realizes a typed extension witness and distinguishes setwise resistance from readout invariance.

20.3 Positive Regime Structure

Definition 20.8 (Positive Regime Space)**.** Let $\mathcal{E}^+ = \mathcal{E} \setminus \{\varnothing_\phi\}$ with inherited order. For a witness pair, let $A = \{\Phi(z_0), \Phi(z_1)\} \setminus \{\varnothing_\phi\}$ and $A_0 = A \cup \{\varnothing_\phi\}$.

Theorem 20.9 (Minimal Positive Witness Regime)**.** *The finite positive witness image A is non-empty and has a minimal element. If its two possible values are comparable, it has a unique least element. No ambient order completeness is needed. This does not prove a least member of all of $\mathcal{E}^+$.*

Proof. Non-emptiness follows from joint null exclusion. A strictly descending chain in a finite non-empty poset terminates at a minimal element. If at most two members are comparable, the smaller is the unique least member. In the complete lattice $\{0, a, b, 1\}$, incomparable positive a, b are both minimal in their two-point image and have infimum zero. Thus an ambient infimum alone gives neither a positive infimum nor uniqueness. □

Definition 20.10 (Structural Intensity)**.** Distinguish null distance $r(e) = d_{\mathcal{E}}(e, \varnothing_\phi)$, global lower gap $\iota(e) = \inf_{f \in \mathcal{E},\ f \prec e} d_{\mathcal{E}}(e, f)$, and the audited relative gap $\iota_{A_0}(e)$ obtained by restricting this infimum to $f \in A_0$. All are measured in units of $d_{\mathcal{E}}$. A positive null distance does not imply a positive global gap.

Theorem 20.11 (Anchored Witness Intensity Bound)**.** *If one endpoint is null, the other image e is the unique positive member of A and satisfies $\iota_{A_0}(e) = r(e) \geq C\epsilon > 0$. Without that anchor, only $\max_i r(\Phi(z_i)) \geq C\epsilon/2$ follows. No universal global intensity bound is claimed.*

Proof. Apply the anchored corollary. The only smaller member of A_0 is null, so the relative infimum equals null distance. On the usual interval $[0, 1]$, the global gap at 1 is zero despite relative gap one in $\{0, 1\}$, showing why the restriction matters. □

Theorem 20.12 (Conditional Phenomenological Closure)**.** *A declared detected pair gives joint null exclusion and a minimal positive member of its finite image. A null anchor gives a unique positive witness value and the anchored quantitative bound. Uniqueness in the ambient regime space, temporal stability and CCR realizability require further hypotheses and constructions.*

Proof. Combine the preceding witness, order and relative-gap results with their stated hypotheses. None of these implications supplies the premises of another by assuming the desired conclusion. □

21 Canonical Families and Structural Closure Consequences

Proposition 21.1 (Profinite Bound)**.** *A profinite witness pair with declared positive separation ϵ, certified lower constant $C_{\mathrm{prof}} > 0$ and one null endpoint gives $r(e) \geq C_{\mathrm{prof}}\epsilon$. Summable coordinate costs do not by themselves certify CCR.*

Proof. Apply the anchored bound. A transformation changing infinitely many coordinates can still cost at most the finite sum of the coordinate weights, so the old finite-support argument cannot establish infinite mass. □

Proposition 21.2 (Symbolic Bound)**.** *The same anchored conclusion holds for a symbolic witness with its audited metric and lower constant. The full binary shift with complement intervention and summable coordinate-change costs is a control against automatic infinite mass.*

Proof. Apply the anchored bound. Complement changes every symbol at finite total summable cost; when admitted it yields positive deformation and finite infimum mass. □

Theorem 21.3 (Rigidity Inheritance)**.** *For $\Phi' = b \circ \Phi \circ a$, lower constants c_a for a, C for Φ on the mapped pairs and c_b for b give lower constant $c_a c_b C$. CCR inheritance separately requires every new finite-cost positive deformation to pull back to one in the old sector. Summability alone proves neither statement.*

Proof. Compose the three inequalities. A new finite-cost positive deformation would contradict original CCR under the stipulated pullback property. Null anchors and stability moduli need their own transport conditions. □

Theorem 21.4 (Factorization on a Faithful Image)**.** *If $\Phi : Z \to \mathcal{E}$ has a positive global lower bound, then every map $\Psi : Z \to F$ factors uniquely through $\Phi(Z)$. If Ψ is L-Lipschitz, the factor is L/C-Lipschitz. No unique extension to the whole ambient $\mathcal{E}$ is implied.*

Proof. The lower bound makes Φ injective with inverse Lipschitz constant $1/C$ on its image. Set $u(\Phi(z)) = \Psi(z)$; this is well-defined, unique there, and has the stated bound. Outside the image no value is prescribed. □

Theorem 21.5 (Finite-Prefix Simulation Bound)**.** *For $F(b) = \sum_{j\geq 0} 2b_j/3^{j+1}$ and a decoder observing only bits $0, \ldots, N$, the optimal worst-case scalar error is $1/(2\,3^{N+1})$. It is positive for every fixed prefix and tends to zero with growing budget. The conclusion concerns prefix information, not finite real dimension.*

Proof. Two extreme continuations of the same prefix have images separated by $3^{-(N+1)}$, forcing half that error for any single decoder value. The midpoint attains the bound on every fiber. This also supplies a non-empty approximation model consistent with revised E1/E2. □

Theorem 21.6 (Conditional Closure by Non-Alternatives)**.** *Any alternative model satisfying the same declared pair, lower-bound, anchor and transport hypotheses obeys the corresponding preceding conclusions. The program of realizing these properties jointly with an infinite-mass intervention model remains open; pair separation is not a proof of an individual null state's dynamical instability.*

Proof. The conclusions depend on the enumerated mathematical hypotheses and apply to every model satisfying them. Existence and physical interpretation are additional questions. □

Part VI
BaseCore Boundary and Expansion Interface

22 BaseCore Ownership and Downstream Boundary

Definition 22.1 (BaseCore Envelope)**.** The BaseCore envelope is the collection of theorem-level structures owned directly by this source tree:

1. foundational Hilbert-space projection dynamics;
2. typed computable witnesses and state spectral-gap bounds;
3. parameterwise non-collapse under explicit anti-constant assumptions;
4. inverse-limit identity under non-finite-determination assumptions;
5. conditional rigidity under admissible perturbation models;
6. conditional obstruction to uniform response fidelity under a binary representation budget;
7. operational-criterion grammar, certification, falsation, and claim-boundary ledgers.

Proposition 22.2 (Boundary Discipline)**.** *BaseCore may define structures that downstream layers use, but it does not claim downstream closure. In particular, BaseCore does not by itself own runtime thresholds, bridge admissibility, human-comparator programs, phenomenality burdens, or Papers 7--9.*

Proof. These excluded layers require additional assumptions, protocols, or empirical burdens not stated in BaseCore. Therefore they cannot be claimed as BaseCore theorems without violating the source-level claim boundary. □

Owned by BaseCore	**Expanded but not owned here**	**Kept downstream**
Projection dynamics, fixed points, compact attractors, anti-constant H5 burden, typed computable witness, inverse-limit identity under NFD, conditional rigidity, conditional non-simulability, operational-criterion grammar, falsation ledger	Non-convex class variants, $\mathscr{C}^R$ style extensions, Ω dynamics, threshold numerics, mutable runtime class ledgers, implementation-specific status surfaces	Paper 7 operational life / subjecthood, Paper 8 indexed subjectivity, Paper 9 phenomenal bridge organization, Stage 2 / Stage 3 comparative programs, human comparator families, phenomenality and external adjudication

Remark 22.3 (Why the earlier material was removed from BaseCore ownership)*.* Previous versions mixed theorem-tight base results with runtime-like numerics, non-convex extension classes, and bridge-facing vocabulary. That blurred the mathematical base layer. BaseCore now keeps only the theorem ownership that survives the stricter release gate.

23 Admissible Structural Extensions

Definition 23.1 (BaseCore export object)**.** A BaseCore export object is a tuple

$$\mathfrak{E} = (\mathcal{O}, \mathcal{H}, \mathcal{R}, \mathcal{B})$$

where $\mathcal{O}$ is a typed mathematical object defined in BaseCore, $\mathcal{H}$ is the finite list of hypotheses under which it is used, $\mathcal{R}$ is the list of results actually proved from those hypotheses, and $\mathcal{B}$ is a boundary list of claims not entailed by $\mathcal{R}$.

Definition 23.2 (Downstream admissible extension)**.** A downstream layer is an admissible extension of a BaseCore export object $\mathfrak{E}$ only if it supplies:

1. an explicit target predicate P not already contained in $\mathcal{R}$;
2. a typed construction or estimator witnessing the objects on which P is evaluated;
3. a proof, reduction, or reproducible protocol connecting $\mathcal{H}$ to P;
4. negative controls or failure modes under which P must not be asserted;
5. a provenance statement distinguishing theorem, implementation, internal support, and external adjudication.

Proposition 23.3 (No automatic downstream promotion)**.** *Let $\mathfrak{E} = (\mathcal{O}, \mathcal{H}, \mathcal{R}, \mathcal{B})$ be a BaseCore export object and let P be a predicate introduced outside BaseCore. If $P \notin \mathcal{R}$, then BaseCore does not entail P unless an admissible extension supplies an additional bridge from $\mathcal{H}$ to P.*

Proof. By definition, $\mathcal{R}$ records the results proved from $\mathcal{H}$ inside BaseCore. A predicate P introduced downstream is not in that result set. Therefore asserting P from BaseCore alone would add an unstated inference. The missing inference must be supplied as a separate bridge, proof, reduction, or reproducible protocol. □

Criterion 23.4 (Runtime-use admissibility)**.** A runtime measurement may instantiate a BaseCore export object only as an implementation-level witness when all of the following are recorded: the measured state space, the estimator or update rule, the finite horizon, the calibration regime, the controls under which the measurement fails, and the claim class being asserted. Such a measurement is not a BaseCore theorem and is not external validation by itself.

Remark 23.5 (Open-load placement for I_{int} and atomic separators)*.* Quantities such as I_{int} and any atomic-separator condition may be treated as downstream target predicates or estimator families only after their carrier objects, separator class, invariance target, and negative controls are specified. Until a proof or reproducible adversarial protocol closes that burden, BaseCore may export the ambient structural language but not the separator conclusion.

Proposition 23.6 (Boundary preservation under finite implementations)**.** *Suppose a finite implementation produces diagnostics for a downstream predicate P over a bounded horizon. If the implementation does not prove that the diagnostics are invariant over the relevant mathematical class, then the diagnostics remain implementation evidence for P and do not promote P to a BaseCore result.*

Proof. The implementation supplies finite observations or computations. Promotion to a BaseCore result requires a class-level implication from the stated hypotheses to the predicate. Without that implication, the observations can support, falsify, or calibrate a downstream claim, but they do not alter the theorem inventory of BaseCore. □

Part VII
Operational-Criterion Grammar and Certification

24 Admissible Systems, Supports, and Histories

24.1 System Model

Definition 24.1 (Admissible system)**.** An admissible system is a tuple

$$S = (X, \Phi, C, R, \Gamma, U)$$

where:

1. X is a metrizable internal state space with metric d_X;
2. $\Phi = \{\Phi_u\}_{u \in U}$ is a family of admissible transition operators $\Phi_u : X \to X$;
3. C is the effective causal structure used to test admissible factorization and intervention behavior;
4. $R = \{r_\alpha\}_{\alpha \in \Lambda_R}$ is a family of readouts, each $r_\alpha : X \to Y_\alpha$;
5. $\Gamma = \{\gamma_j\}_{j=1}^{m}$ is an admissibility bundle defining
$$X_\Gamma = \{x \in X : \gamma_j(x) \le 0 \ \forall j\};$$
6. U is the set of admissible interventions.

Definition 24.2 (Declared State Attractor)**.** Fix a non-empty compact set $\mathrm{Attr}_X(S) \subseteq X_\Gamma$ invariant under every admitted transition and a non-empty class of admitted schedules at every horizon. It is a controlled attracting region on a declared non-empty basin B_X if, for every $x_0 \in B_X$,

$$\sup_{u_\bullet} \mathrm{dist}_{d_X}(x_n, \mathrm{Attr}_X(S)) \longrightarrow 0, \qquad x_{k+1} = \Phi_{u_k}(x_k).$$

The supremum is over admitted schedules. This is a state-space certificate to supply, not an existence theorem. Its codomain is X; the structurally stable readout set $\mathrm{Attr}(S)$ in the null-regime section instead lies in $\mathcal{E}$. No intersection of those different types is taken.

24.2 Admissible Support

Definition 24.3 (Admissible support)**.** A subset $A_S \subseteq X_\Gamma$ is an admissible support if:

1. A_S is non-empty and compact;
2. $\Phi_u(A_S) \subseteq A_S$ for every $u \in U$;
3. $A_S \cap \mathrm{Attr}_X(S) \neq \varnothing$;
4. A_S stays disjoint from the collapse set excluded by the base non-collapse burden.

24.3 Operational Histories

Definition 24.4 (Windowed readout history)**.** Fix a horizon $T \in \mathbb{N}$ and an intervention schedule $u_\bullet = (u_0, \dots, u_{T-1})$. For $x \in A_S$, define

$$h_{S,T}(x; u_\bullet) = \big(r_\alpha(x_t)\big)_{0 \le t \le T,\ \alpha \in \Lambda_R}, \qquad x_{t+1} = \Phi_{u_t}(x_t), \quad x_0 = x.$$

The set of all such histories is denoted $\mathscr{H}_{S,T}$.

Definition 24.5 (Operational partition candidate)**.** An operational partition candidate is a map

$$\pi_{S,T} : \mathscr{H}_{S,T} \to \mathscr{Q}_{S,T}$$

into a finite or countable label set $\mathscr{Q}_{S,T}$.

24.4 Witness Margin Vector

Definition 24.6 (Witness margin vector)**.** Define the witness margin vector

$$\Delta(S) = \big(\delta_{\mathrm{per}}(S), \delta_{\mathrm{ri}}(S), \delta_{\mathrm{int}}(S), \delta_{\mathrm{cont}}(S), \delta_{\mathrm{diff}}(S), \delta_{\mathrm{leg}}(S)\big) \in [0, \infty)^6.$$

When the six witnesses are separately calibrated as certified radii in the same perturbation metric and class, their invariant budget is

$$\delta_\star(S) = \min \Delta(S).$$

A positive radius certifies the associated predicate at the target and throughout its declared perturbation ball. A zero or unreported radius alone does not determine failure at the target. Raw separations, risks, durations, fragmentation ceilings and compression errors require separate calibration before taking this minimum.

Invariant	Certified content	Upstream anchor	Rupture signature
I_{per}	Persistent admissible support	BaseCore dynamics	Support loss or collapse exposure
I_{ri}	Unique rigid identity object on support	Identity / rigidity block	Identity ambiguity
I_{int}	No admissible non-trivial factorization	Criterion grammar	Reducible aggregate
I_{cont}	Regime continuity under admissible evolution	Regime constraints	Fragmentation into incompatible classes
I_{diff}	Stable exclusion of null or trivial collapse	Null-regime exclusion	Null or trivialized class structure
I_{leg}	Decoder-certified recoverability under controls	Falsation grammar	Opaque or non-auditable partition

25 Six Critical Invariants

25.1 Structural Persistence

Definition 25.1 (Structural persistence)**.** An admissible system satisfies structural persistence, written $I_{\mathrm{per}}(S) = 1$, if there exists an admissible support A_S and a constant $\delta_{\mathrm{per}}(S) > 0$ such that:

1. A_S is forward invariant under all $u \in U$;
2. every admissible trajectory starting in A_S remains at least $\delta_{\mathrm{per}}(S)$ away from the collapse set;
3. A_S contains or converges into a non-empty admissible attractor region.

25.2 Rigid Identity

Definition 25.2 (Rigid identity)**.** An admissible system satisfies rigid identity, written $I_{\mathrm{ri}}(S) = 1$, if the observable family induced by R on A_S determines a unique identity object $\mathcal{I}_S$ and there exists a gap $\delta_{\mathrm{ri}}(S) > 0$ such that every admissible alternative identity candidate remains at weighted deformation distance at least $\delta_{\mathrm{ri}}(S)$ from $\mathcal{I}_S$.

25.3 Causal Integration

Definition 25.3 (Causal integration)**.** An admissible system satisfies causal integration, written $I_{\mathrm{int}}(S) = 1$, if there is no non-trivial factorization

$$A_S = A_1 \times A_2, \qquad R = R_1 \sqcup R_2,$$

together with decomposed dynamics and causal structure reproducing admissible histories within error smaller than a positive margin $\delta_{\mathrm{int}}(S)$ while preserving $\mathcal{I}_S$. More explicitly, this shorthand is relative to a declared admissible factorization class. For such a class $\mathfrak{D}$, write $I_{\mathrm{int}}(S;\mathfrak{D}) = 1$ when no factorization in $\mathfrak{D}$ reproduces the admissible histories below a positive margin $\delta_{\mathrm{int}}(S;\mathfrak{D})$ while preserving $\mathcal{I}_S$. The unqualified notation $I_{\mathrm{int}}(S) = 1$ refers to the default structural class $\mathfrak{D}_*$: fixed, time-homogeneous, schedule-independent, non-trivial product factorizations with both factors non-singleton, with split readouts $R = R_1 \sqcup R_2$, decomposed dynamics and causal structure, a search space normalized independently of the target error, and no reconstruction decoder or time-dependent post-processing map that recouples the factors at output time. This default structural reading is the adopted class recorded in `docs/ai-platform-outputs/analysis/QICN_IINT_CANONICAL_CLASS_SPLIT_READOUT.md`.

Remark 25.4 (Class-relative integration margin)*.* The integration criterion is class-relative. Adopting $\mathfrak{D}_*$ does not assert that S resists every broader behavioral simulator. In particular, under decoder-coupled or schedule-dependent behavioral classes such as $\mathfrak{D}_{\mathrm{approx}}$, the corresponding margin $\delta_{\mathrm{int}}(S;\mathfrak{D}_{\mathrm{approx}})$ may vanish, i.e. may be equal to 0; the current coupled-carrier analysis records precisely this failure mode. Thus the default structural reading preserves a falsation surface and must not be read as an unconditional closure of integration, nor as a strengthening of any downstream theorem that assumes $I_{\mathrm{int}}(S) = 1$.

Lemma 25.5 (Two Factor Fibers Cannot Share a Nonconstant Label)**.** *Let $\chi : A \to A_1 \times A_2$ be surjective. If a label $g : A \to L$ is constant on every fiber of each of the two factor projections of χ, then g is constant on A.*

Proof. For any $x, y \in A$, choose z whose first factor is that of x and whose second factor is that of y. Fiber constancy gives $g(x) = g(z) = g(y)$. For a quantitative obstruction it suffices that the proposed history error below a fixed threshold implies both fiber-constancy statements for a nonconstant label. Non-emptiness of the comparator sector is a separate condition. □

25.4 Regime Continuity

Definition 25.6 (Regime continuity)**.** An admissible system satisfies regime continuity, written $I_{\mathrm{cont}}(S) = 1$, if there exists a complete metric space $(\mathcal{E}, d_{\mathcal{E}})$ and an admissible assignment

$$\Pi_S : A_S \to \mathcal{E}$$

such that:

1. every admissible trajectory induces a continuous path in $\mathcal{E}$ on admissible windows;
2. there exists $\delta_{\mathrm{cont}}(S) > 0$ bounding the fragmentation functional on all admissible histories;
3. no admissible finite-energy perturbation produces a discontinuous jump into an incompatible regime class while $\mathcal{I}_S$ is preserved.

25.5 Non-Null Differentiation

Definition 25.7 (Non-null differentiation)**.** An admissible system satisfies non-null differentiation, written $I_{\mathrm{diff}}(S) = 1$, if there exist $x, y \in A_S$, a horizon T, and a readout subfamily $\mathscr{R}^\star \subseteq R$ such that

$$\operatorname{dist}(h_{S,T}(x; u_\bullet), h_{S,T}(y; u_\bullet)) \geq \delta_{\mathrm{diff}}(S) > 0$$

for some admissible schedule $u_\bullet$, and if the null class $\mathbf{0}$ is not an asymptotically stable admissible attractor.

25.6 Operational Legibility

Definition 25.8 (Decoder Family and Structural Closure)**.** A decoder family uses one declared class set and acts on the specified history windows. For each intervention-response comparison, declare its response cells and a matching relation on those cells. The family is structurally closed when decoder class equality composes exactly and the response-cell matching relations are reflexive, symmetric and transitive. These are conditions on the chosen comparison data; they are not inferred from a numerical tolerance alone. The legibility clauses below supply the additional operational requirements.

Definition 25.9 (Operational legibility)**.** An admissible system satisfies operational legibility, written $I_{\mathrm{leg}}(S) = 1$, if there exist:

1. a declared readout subfamily $\mathscr{R}^{\mathrm{leg}} \subseteq R$;
2. a structurally closed decoder family $\mathscr{D}_S$ acting on windows of length at least $\tau_{\min}(S)$;
3. a class set $\mathscr{Q}_S$;
4. a measurement protocol with separation $\sigma > 0$, readout noise radius $r > 0$, minimum duration $\tau_{\min} \geq 1$, miss and false-positive budgets $p_{\mathrm{miss}}, p_{\mathrm{fp}} \in [0, 1)$, and a finite compression budget B with class-distortion tolerance d_{comp};

such that:

(L1) Separability Distinct declared classes have distance at least σ in the specified clean comparison domain.

(L2) Noise robustness On the specified noise domain, perturbations with readout norm at most r preserve the label.

(L3) Persistence window Under declared sham schedules, labels remain stable throughout every tested window of duration $\tau_{\min}$.

(L4) Intervention fidelity In a declared nonempty target population, the miss probability for a predicted class shift is at most p_{miss}.

(L5) Negative controls In a declared nonempty sham population, the false-shift probability is at most p_{fp}.

(L6) Structured compressibility An encoder with at most 2^B codes and a code-to-class map preserve nontrivial class structure with distortion at most d_{comp} in the declared class metric.

Proposition 25.10 (Each invariant is individually necessary)**.** *If any one of* $I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}$ *fails, then* $S \notin \mathrm{Crit}_{\mathrm{op}}$.

Proof. Without I_{per} there is no stable admissible domain. Without I_{ri} there is no unique identity carrier. Without I_{int} the system is reducible. Without I_{cont} the criterion fragments under admissible evolution. Without I_{diff} the admissible partition collapses into nullity or triviality. Without I_{leg} the partition is not recoverable under admissible readout and intervention discipline. Since Definition 26.6 below requires all six, failure of any one excludes membership. □

26 Criterion Class, Certified Partition, and Rupture

Definition 26.1 (Certified decoder)**.** A decoder $D \in \mathscr{D}_S$ is certified as part of a structurally closed family when the six clauses of Definition 25.9 hold on the chosen admissible support. Structural closure and operational certification are separate requirements.

Definition 26.2 (Criterion indistinguishability)**.** For $x, y \in A_S$, write $x \sim_{\mathrm{Part}_{\mathrm{op}},S} y$ if, for every certified decoder $D \in \mathscr{D}_S$, every admissible horizon $T \geq \tau_{\min}(S)$, and every admissible intervention schedule $u_\bullet$,

$$D(h_{S,T}(x; u_\bullet)) = D(h_{S,T}(y; u_\bullet)),$$

and the intervention responses remain matched within the legibility margin.

Proposition 26.3 (Well-definedness of the certified partition)**.** *On any declared support A_S with a structurally closed decoder family, the relation $\sim_{\mathrm{Part}_{\mathrm{op}},S}$ is an equivalence relation. This conclusion does not require the other critical invariants; certification of criterion membership still does.*

Proof. For every fixed decoder, horizon and intervention schedule, equality of decoder labels is reflexive, symmetric and transitive. The response-cell matching relation has those properties by Definition 25.8. Their intersection, over all declared comparisons, has the same properties: if x matches y and y matches z, each decoder equality and each response-cell match composes, so x matches z. This proves the assertion without adding two tolerance errors and silently treating the sum as the original tolerance. □

Remark 26.4 (Tolerance Alone Is Not Transitive)*.* With constant decoder labels, let response matching mean distance at most 1 on $\{0, 3/4, 3/2\}$. Adjacent points match, but the endpoints do not. Thus decoder equality plus raw tolerance is not enough. By contrast, equality under a declared response-cell map is an equivalence relation and supplies a concrete structurally closed matching rule. This example concerns closure only, not satisfaction of all six legibility clauses.

Definition 26.5 (Certified operational partition)**.** Assume Proposition 26.3. The certified operational partition of S is

$$\mathrm{Part}_{\mathrm{op}}(S) := A_S/{\sim_{\mathrm{Part}_{\mathrm{op}},S}}.$$

Definition 26.6 (Operational-criterion class membership)**.** An admissible system belongs to the BaseCore operational-criterion class, written $S \in \mathrm{Crit}_{\mathrm{op}}$, if there exists a non-empty admissible support A_S such that

$$I_{\mathrm{per}}(S) = I_{\mathrm{ri}}(S) = I_{\mathrm{int}}(S) = I_{\mathrm{cont}}(S) = I_{\mathrm{diff}}(S) = I_{\mathrm{leg}}(S) = 1.$$

Remark 26.7 (Neutrality of the class symbol)*.* The symbol $\mathrm{Crit}_{\mathrm{op}}$ is a shorthand for a conjunctive operational criterion. It is not a theorem of phenomenality, metaphysical subjectivity, or human equivalence.

Definition 26.8 (Rupture)**.** A rupture is any loss of admissibility or loss of at least one critical invariant margin beyond its admissible threshold.

Theorem 26.9 (Rupture by Invariant Loss)**.** *If a critical invariant has no admissible witness on any support, then $S \notin \mathrm{Crit}_{\mathrm{op}}$. Loss of a witness on one chosen support invalidates that support's certificate, but does not exclude a certificate on another support without further analysis. Quotient undefinedness, triviality and loss of legibility are not mutually exclusive labels and need not accompany every loss of criterion membership.*

Proof. A criterion witness requires all six invariants on one admissible support. Global failure of one invariant rules out such a support. A certificate on a fixed support must satisfy every clause there, so a local witness loss invalidates that certificate. A structurally closed decoder relation may still define a quotient when another invariant, such as integration, fails; the quotient theorem does not assume that invariant. Hence an exclusive three-way classification cannot be inferred. □

27 Minimality of the Invariant Basis

Definition 27.1 (Reduced criterion)**.** Let

$$\mathfrak{I} = \{I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}\}.$$

For every subset $J \subseteq \mathfrak{I}$ define the reduced class

$$\mathcal{C}_{\mathrm{op}}(J) := \{S : I(S) = 1 \text{ for every } I \in J\}.$$

Definition 27.2 (Single-rupture witness family)**.** Fix one invariant index $k \in \{\mathrm{per}, \mathrm{ri}, \mathrm{int}, \mathrm{cont}, \mathrm{diff}, \mathrm{leg}\}$. A single-rupture witness family $\mathcal{W}_k$ consists of admissible systems for which predicate I_k fails and every other predicate holds on the declared comparison domain.

Proposition 27.3 (Single-rupture witnesses create false positives)**.** *Let $J_k = \mathfrak{I} \setminus \{I_k\}$ and assume $\mathcal{W}_k \neq \varnothing$. Then every $S \in \mathcal{W}_k$ satisfies*

$$S \in \mathcal{C}_{\mathrm{op}}(J_k) \qquad \textit{but} \qquad S \notin \mathrm{Crit}_{\mathrm{op}}.$$

Proof. All retained invariants remain positive by construction, but the omitted invariant fails exactly. Proposition 25.10 then excludes S from $\mathrm{Crit}_{\mathrm{op}}$. □

Theorem 27.4 (Minimality of the six-invariant basis)**.** *Let $\mathfrak{U}$ be any admissible universe containing at least one single-rupture witness for each invariant. Then no strict subset of $\mathfrak{I}$ is extensionally equivalent to* $\mathrm{Crit}_{\mathrm{op}}$ *on $\mathfrak{U}$.*

Proof. Take any strict subset $J \subsetneq \mathfrak{I}$. Some invariant I_k is omitted. By assumption, $\mathfrak{U}$ contains a witness $S_k \in \mathcal{W}_k$, which satisfies the reduced criterion but not $\mathrm{Crit}_{\mathrm{op}}$. Hence the two classes differ on $\mathfrak{U}$. □

28 Operational Legibility as a Certified Object

Criterion 28.1 (Legibility certificate)**.** An operational partition candidate is certified only if:

1. comparator separability holds against pre-registered baselines;
2. admissible noise does not dissolve the class map below the legibility margin;
3. the same class assignment survives on windows of length at least $\tau_{\min}(S)$;
4. targeted interventions on an invariant induce the predicted directional class change;
5. sham or off-target interventions do not generate the same class signature;
6. audit-friendly compression preserves class structure without total collapse.

Proposition 28.2 (Legibility is jointly constrained)**.** *No single legibility metric is sufficient for I_{leg}. The invariant requires simultaneous control of separation, admissible noise robustness, persistence window, intervention fidelity, negative controls, and structured compressibility.*

Proof. A decoder can be separable yet fragile to admissible noise, stable over windows yet intervention-insensitive, or robust under noise yet collapse under compression. Because Definition 25.9 is conjunctive, all six clauses are required together. □

29 Certified Candidate Audit Schema

Definition 29.1 (Base criterion certificate)**.** A BaseCore criterion certificate for an admissible system S is a tuple

$$\mathrm{Cert}(S) = (A_S, \mathscr{D}_S, \pi_S, \Delta(S), E_S),$$

where:

1. A_S is a certified admissible support;
2. $\mathscr{D}_S$ is a certified decoder family;
3. π_S is the decoder-induced partition on admissible histories;
4. $\Delta(S)$ is the witness margin vector;
5. E_S is the evidence bundle localizing the witness for each invariant.

Criterion 29.2 (Candidate certification rule)**.** A candidate system may be certified as a member of $\mathrm{Crit}_{\mathrm{op}}$ only if all of the following are available on the same admissible support:

1. a non-empty compact forward-invariant support certificate;
2. a positive witness margin vector $\Delta(S) > 0$ coordinate-wise;
3. a decoder family satisfying Definition 25.9;
4. a targeted intervention panel together with negative controls;
5. an auditable compression map preserving class structure;
6. a traceable dependency ledger showing which upstream result supports each invariant witness.

Theorem 29.3 (Certified criterion membership)**.** *If* $\mathrm{Cert}(S)$ *exists and satisfies Criterion 29.2, then* $S \in \mathrm{Crit}_{\mathrm{op}}$ *and* $\mathrm{Part}_{\mathrm{op}}(S)$ *is well-defined.*

Proof. Criterion 29.2 explicitly requires positive witnesses for all six invariants on a common admissible support. Definition 26.6 then gives $S \in \mathrm{Crit}_{\mathrm{op}}$, and Proposition 26.3 gives a well-defined partition quotient. □

Remark 29.4 (Boundary discipline). The section above defines a theorem-tight grammar for certification, rupture, and falsation. It does not prove phenomenality, subjectivity, biological primacy, or cross-substrate identity transport.

Part VIII
Claim Boundary and Falsation Grammar

30 Forensic Admissibility and Integrity Grammar

Definition 30.1 (Forensic packet)**.** Each operational run emits a packet

$$K = (\mathrm{tick}, \mathrm{seed}, \mathrm{versionHash}, H_v, \mathrm{metrics}, \mathrm{sensorSummary}, \mathrm{checksum}).$$

Assumption 30.2 (Packet admissibility)**.** A run is admissible only if checksum, schema, seed consistency, version hash, and H_v consistency all pass.

Definition 30.3 (Severe violation)**.** A run has a severe violation if any of the following holds: checksum mismatch, schema failure, seed/version inconsistency, or critical-state replay mismatch.

Criterion 30.4 (Integrity gate)**.** Severely violated runs are excluded from support counting and marked as hard failures.

Proposition 30.5 (Integrity-first inference)**.** *If admissibility fails, inferential significance is non-actionable.*

31 Certification, Comparison, and Rupture

Definition 31.1 (Operational Certification Indicator)**.** Keep $\mathrm{Cert}(S)$ for the certificate object of Definition 29.1. Define a distinct mathematical indicator $\mathrm{CertFlag}(S)$ to be 1 exactly when such a certificate satisfies Criterion 29.2 on one common admissible support and meets the integrity grammar above; otherwise its value is 0. This does not identify a tuple with a Boolean value. An empirical audit may have an unknown verdict: missing evidence is not a proof that this mathematical indicator is zero.

Definition 31.2 (Equivalence discriminator)**.** For two admissible systems S_1, S_2, define

$$\mathrm{Eq}_{\Gamma,\varepsilon}(S_1, S_2) = 1$$

when the relevant signatures, admissibility bundle, decoder behavior, and invariant deltas match within the allowed comparison regime.

Definition 31.3 (Rupture event)**.** For a targeted invariant family J, define

$$\mathrm{Rupt}_J(S) = 1$$

when an admissible intervention or perturbation drives at least one invariant in J below its certification margin so that class membership becomes invalidated, destroyed, or undefined.

32 Prediction and Falsation Grammar

Proposition 32.1 (Complexity-only negative control prediction)**.** *If a candidate exhibits high connectivity, high entropy, or rich activity but no admissible support realizes all required critical witnesses together, then*

$$\mathrm{CertFlag}(S) = 0, \qquad S \notin \mathrm{Crit}_{\mathrm{op}} \text{ in certified form}, \qquad \mathrm{Part}_{\mathrm{op}}(S) \text{ is not certified.}$$

Proposition 32.2 (Ablation prediction)**.** *If an intervention destroys a required witness on the certified support, that certificate loses validity. Excluding a replacement certificate on a different support requires checking that support as well.*

Proposition 32.3 (Exact grammar-transport prediction)**.** *If two realizations preserve the relevant signatures, admissibility conditions, decoder behavior, and six critical invariants up to the allowed comparison regime, then the framework predicts preserved BaseCore criterion membership.*

Proposition 32.4 (Threshold-sensitive boundary behavior)**.** *The framework predicts stable pass/fail regions together with narrow transition bands near critical margins. It predicts neither global threshold collapse nor perfect flat insensitivity.*

Proposition 32.5 (Legibility dependence prediction)**.** *Criterion membership becomes inadmissible if readouts cease to be separable, noise-robust, intervention-sensitive, or compressible without total collapse.*

Proposition 32.6 (Continuity and non-null relevance)**.** *Regime continuity and non-null differentiation are constitutive. Systems that preserve rich activity while losing continuity or collapsing toward effective null structure fail the operational criterion or leave it undefined.*

Remark 32.7 (What stays outside BaseCore)*.* Runtime status classes, Stage 2 / Stage 3 comparative programs, human comparator campaigns, Papers 7--9, and bridge-facing burden architectures remain downstream artifacts. BaseCore absorbs only the falsation grammar and claim-boundary semantics needed to keep theorem ownership honest.

33 Claim Boundary Ledger

33.1 Allowed BaseCore Claims

1. Under H1--H4, projection dynamics admit unique fixed points and compact attractor families.
2. Under H5, attractors are parameterwise non-constant.
3. Under NFD, inverse-limit identity is not determined by any finite canonical projection.
4. Under RIG-1 through RIG-5, inverse-limit structures are stable under summable admissible perturbations in the weighted metric.
5. Under NS-1 through NS-3, a binary budget bounds the number of separated responses that can be recovered at uniform tolerance.
6. Under the six-invariant operational burden, certified criterion structures can be defined, audited, and falsified.

33.2 Forbidden BaseCore Claims

1. Human phenomenal consciousness.
2. Metaphysical subjectivity.
3. Human-machine equivalence.
4. External validation.
5. Runtime instantiation by the current system.
6. Paper 9 bridge admissibility.
7. Strong phenomenality.
8. Moral status.
9. Cross-substrate identity preservation.
10. Public claim closure.

34 Dependency Ledger

1. The foundational Hilbert-space dynamics remain the base layer.
2. Papers 1--6 contribute formal material selectively normalized into BaseCore only where theorem ownership belongs in the base.
3. Papers 7--9 remain downstream and are not dependencies of any BaseCore theorem.

4. Stage 2 / Stage 3 comparative programs remain downstream runtime and governance layers.

5. The PDF remains a derived artifact of the LaTeX tree.

35 Assumption-to-Theorem-to-Non-Claim Mapping

Assumption block	**Theorem family**	**Non-claim boundary**
H1--H4	Projection existence, contraction, unique fixed points, compact attractor family	No parameterwise non-collapse and no identity / rigidity / simulator claim
H5	Structural non-collapsibility	No claim about phenomenality, subjectivity, or runtime class membership
NFD	Non-locality of identity under canonical finite projections	No theorem that arbitrary set-theoretic injections into finite slices are impossible
RIG-1 through RIG-5	Conditional stability of inverse-limit identity and weighted Hausdorff control	No automatic isomorphism, shape equivalence, or transfer theorem without extra assumptions
NS-1 through NS-3	Binary-budget obstruction and sharp prefix error	Real dimension and intervention cost are different resources
Six invariant burden plus legibility clauses	Criterion class membership, certified partition, rupture logic	No theorem of phenomenality, human equivalence, biology as primitive, or bridge admissibility

36 Theorem Hygiene Ledger

Theorem	Assumptions	Conclusion type	Non-conclusion	Downstream use
Existence and uniqueness of projection	H1	Hilbert-space geometric existence / uniqueness	No empirical or runtime content	Base transition grammar
Contractivity	H1--H2	Strict contraction bound	No spectrum with unit state eigenvalue	Fixed-point and spectral-gap results
Unique fixed point	H1--H3	Existence and convergence	No non-collapse by itself	Attractor family
Compactness of attractor family	H1--H4	Compactness / continuity	No regime or identity burden by itself	Parameter-family base layer
Structural non-collapsibility	H1--H5	Parameterwise exclusion of constant fixed points	No stronger semantic interpretation	Anti-collapse gate
State spectral gap	Typed differentiability at fixed point	Spectral-radius upper bound < 1	No persistence mode $\lambda = 1$ in state spectrum	Stability accounting
Non-locality under NFD	Topological regularity plus NFD	Canonical finite slices do not determine identity	No universal ban on arbitrary injections	Identity block
Conditional stability of inverse-limit identity	RIG-1 through RIG-5	Weighted Hausdorff stability under admissible perturbations	No automatic isomorphism	Rigidity accounting
Conditional non-simulability	NS-1 through NS-3	Separated response families require enough codes	Requires a declared budget, observation family and tolerance	Simulator boundary
Certified criterion membership	Six invariants plus certificate rule	Base criterion class and certified partition	No phenomenality or human-equivalence theorem	Downstream classification gates

37 Redundancy Ledger

1. The previous state-spectrum claim $\lambda_1 = 1$ under strict contraction has been removed.
2. The previous untyped Golden Mean witness has been replaced by a typed model with explicit embedding.
3. The earlier non-collapse quantifier mismatch has been removed by replacing the old H5 with a parameterwise anti-constant hypothesis.
4. The earlier absolute inverse-limit nonlocality claim has been replaced by an NFD-conditional theorem.
5. The earlier unconditional rigidity / non-simulability language has been replaced by explicit assumption blocks.
6. Substrate invariance, biological non-primacy, bridge-facing closure, and other downstream claims have been removed from BaseCore ownership.

38 Open Assumptions Inventory

1. The anti-collapse witness is abstract and theorem-valid, but not the same object as the Golden Mean typed witness.
2. Physical realization of CCR-like targets remains open.
3. Empirical instantiation of the BaseCore operational criterion remains open.
4. External validation remains open.
5. Papers 7--9 may build on BaseCore, but their additional burdens remain open at the BaseCore level.

39 Downstream Boundary Note

Papers 7, 8, and 9 remain downstream of this source tree. Their classificatory, indexed-subjectivity, and bridge-burden layers may cite BaseCore, but they are not part of the autonomous mathematical base rebuilt here.

Part IX
Discrete-to-Continuous Bridge

40 From Node Networks to Hilbert Space Dynamics

Boundary of this bridge. This part states a conditional approximation theorem between a finite node-network transition and the abstract BaseCore transition operator from Definition 5.1. It does not prove that any particular runtime execution belongs to an operational-consciousness class, and it does not convert finite tests into external validation.

Definition 40.1 (Discrete State Space)**.** Let $N \geq 1$ and let a node network be indexed by $i \in \{1, \ldots, N\}$. Each node carries:

- a local coherence state $x_i(t) \in [0, 1]$;
- an energy coordinate $e_i(t) \in [0, 1]$;
- a phase coordinate $\phi_i(t) \in [0, 2\pi)$.

The discrete state space is

$$\mathcal{S}_N = [0, 1]^N \times [0, 1]^N \times [0, 2\pi)^N.$$

Definition 40.2 (Mean Coherence Observable)**.** For a discrete state $s(t) = ((x_i(t))_i, (e_i(t))_i, (\phi_i(t))_i) \in \mathcal{S}_N$, define

$$X_N(t) := \frac{1}{N} \sum_{i=1}^{N} x_i(t) \in [0, 1].$$

This is an observable of the coherence coordinate only; it is not a complete state descriptor.

Definition 40.3 (Finite Hilbert Embedding)**.** Let

$$\mathcal{H}_N := \mathbb{R}^N, \qquad \langle \mathbf{a}, \mathbf{b} \rangle_N := \frac{1}{N} \sum_{i=1}^{N} a_i b_i.$$

The coherence embedding

$$\iota_N : \mathcal{S}_N \to \mathcal{H}_N$$

is

$$\iota_N((x_i)_i, (e_i)_i, (\phi_i)_i) = (x_1, \dots, x_N).$$

Thus ι_N forgets energy and phase except where they enter the transition rule below.

Definition 40.4 (Piecewise-Constant Isometry)**.** Let $\mathcal{H} = L^2([0,1])$ and partition $[0,1]$ into intervals

$$I_{N,i} = \left[\frac{i-1}{N}, \frac{i}{N}\right), \qquad i = 1, \dots, N.$$

Define $J_N : \mathcal{H}_N \hookrightarrow \mathcal{H}$ by

$$(J_N \mathbf{x})(s) = x_i \quad \text{for } s \in I_{N,i}.$$

Then J_N is an isometry from $(\mathcal{H}_N, \|\cdot\|_N)$ into $L^2([0,1])$. Its adjoint $J_N^\dagger : \mathcal{H} \to \mathcal{H}_N$ is the cell-averaging map

$$(J_N^\dagger f)_i = N \int_{I_{N,i}} f(s)\, ds.$$

Definition 40.5 (Discrete Transition Operator)**.** Fix auxiliary energy and phase coordinates $(e_i)_i$ and $(\phi_i)_i$, a graph with incoming neighborhoods $\mathrm{conn}(i)$, and a control $u \in \mathcal{U}$. Write

$$C_N = [0,1]^N, \qquad d_N^{\mathrm{in}} = \max_i |\operatorname{conn}(i)|, \qquad d_N^{\mathrm{out}} = \max_j |\{i : j \in \operatorname{conn}(i)\}|.$$

The coherence-coordinate transition

$$\widehat{T}_{N,u}^{e,\phi} : C_N \to C_N$$

is defined componentwise by

$$\left(\widehat{T}_{N,u}^{e,\phi}\mathbf{x}\right)_i = \mathrm{clamp}_{[0,1]}(x_i + \eta R_i(\mathbf{x}; e, \phi) + \alpha\, B_i(\mathbf{x}) + \Gamma_{N,i}(u)),$$

where

$$R_i(\mathbf{x}; e, \phi) = \sum_{j \in \operatorname{conn}(i)} I_j(x_j) \frac{\exp(-|e_i - e_j|/(kT))}{\theta \cos(\phi_i - \phi_j)}.$$

Here $I_j : [0,1] \to [0,1]$ is the intensity readout, B_i is a feedback term, $\Gamma_N(u)$ is the finite input coupling, and $k, T, \theta > 0$, $\eta, \alpha \geq 0$ are parameters. Energy and phase are frozen in this operator; contractivity of this coordinate map does not establish contractivity of a coupled energy--phase evolution.

Assumption 40.6 (Discrete Regularity Conditions)**.** For a fixed parameter region, assume:

1. I_j is L_I-Lipschitz for every node j.
2. The phase denominator is uniformly separated from zero: there exists $c_\phi > 0$ such that $|\cos(\phi_i - \phi_j)| \geq c_\phi$ on every active edge.
3. There exist $\beta > 0$ and an L_D-Lipschitz map $D : C_N \to \mathcal{H}_N$ such that $B(\mathbf{x}) = -\beta \mathbf{x} + D(\mathbf{x})$.
4. $\Gamma_N(u)$ is independent of $\mathbf{x}$ for fixed u.
5. The parameters satisfy

$$L_R := \frac{L_I \sqrt{d_N^{\mathrm{in}} d_N^{\mathrm{out}}}}{\theta c_\phi}, \qquad \rho_N := |1 - \alpha\beta| + \eta L_R + \alpha L_D < 1.$$

Remark 40.7 (Relation to the runtime guard). A denominator guard can ensure that the quotient is defined on a parameter region. It does not establish the dissipative feedback condition or contractivity. On the stated energy domain and for $kT > 0$, the exponential coefficient is at most 1. The conditions above are sufficient, not necessary; a specified feedback must be checked against them.

Lemma 40.8 (Directed Resonance Bound)**.** *Under the first two regularity conditions, R is L_R-Lipschitz in the normalized Hilbert norm.*

Proof. Let a_{ij} be the energy--phase coefficient on an active edge and zero otherwise, and put $W = (|a_{ij}|)$. Componentwise,

$$|R_i(\mathbf{x}) - R_i(\mathbf{y})| \le L_I \sum_j W_{ij}|x_j - y_j|.$$

The row and column sums of W are bounded by $d_N^{\text{in}}/(\theta c_\phi)$ and $d_N^{\text{out}}/(\theta c_\phi)$. The matrix inequality $\|W\|_2 \le \sqrt{\|W\|_1 \|W\|_\infty}$ therefore gives the result. Multiplying both vector norms by $N^{-1/2}$ leaves the matrix bound unchanged. A bound on incoming degree alone does not control this norm for directed graphs. □

Theorem 40.9 (Discrete Operator Contractivity)**.** *Under Assumption 40.6, for fixed u, e, ϕ the operator $\widehat{T}_{N,u}^{e,\phi}$ is a strict contraction on the complete set C_N:*

$$\left\|\widehat{T}_{N,u}^{e,\phi}(\mathbf{x}) - \widehat{T}_{N,u}^{e,\phi}(\mathbf{y})\right\|_N \le \rho_N \|\mathbf{x} - \mathbf{y}\|_N \qquad \forall \mathbf{x}, \mathbf{y} \in C_N.$$

It has a unique fixed point in C_N, and every iterate starting there converges to it.

Proof. The expression before clamping is

$$(1 - \alpha\beta)\mathbf{x} + \eta R(\mathbf{x}) + \alpha D(\mathbf{x}) + \Gamma_N(u).$$

Subtract its values at two states, use Lemma 40.8 and the Lipschitz bound for D, and apply non-expansiveness of the coordinate clamp. This gives the stated factor $|1 - \alpha\beta| + \eta L_R + \alpha L_D$. The input cancels because u is fixed. The cube is nonempty, closed and complete, and the clamp maps into it, so Banach's theorem applies. □

Example 40.10 (Zero-Increment Negative Control). If $R = B = \Gamma_N = 0$, the written transition is the identity on C_N and has Lipschitz constant 1 for $N \ge 1$. It is not a strict contraction. In a decomposition $B = -\beta \,\mathrm{Id} + D$, one has $D = \beta \,\mathrm{Id}$, so $|1 - \alpha\beta| + \alpha\beta \ge 1$ and the sufficient conditions correctly reject it. Lipschitz regularity of increments alone cannot remove the identity term.

Assumption 40.11 (Bridge Consistency)**.** Let $T_u = \aleph(K \cdot + \Gamma(u))$ be the BaseCore transition on $\mathcal{H} = L^2([0,1])$, with $\kappa := \|K\| < 1$. Define the closed invariant-domain candidate

$$C = \{f \in L^2([0,1]) : 0 \le f \le 1 \text{ almost everywhere}\}.$$

Assume $T_u(C) \subseteq C$ and let $F_{N,u} = J_N \widehat{T}_{N,u}^{e,\phi} J_N^\dagger : C \to C$. There is a stated nonnegative bound ε_N such that

$$\sup_{f \in C} \|F_{N,u}(f) - T_u(f)\|_{\mathcal{H}} \le \varepsilon_N.$$

This is a discrepancy between nonlinear maps on a specified domain, not a linear operator norm. Its validity and any claimed rate must be proved for the discretization under consideration; H1--H5 alone do not provide a quadrature rate.

Theorem 40.12 (Hilbert--Discrete Bridge)**.** *Assume Hypotheses 3.1--3.5 and Assumptions 40.6 and 40.11. For $f_0 \in C$ and $\mathbf{x}_0 \in C_N$, set $f_{n+1} = T_u(f_n)$ and $\mathbf{x}_{n+1} = \widehat{T}^{e,\phi}_{N,u}(\mathbf{x}_n)$ with fixed u, e, ϕ. Then, for every $n \geq 0$,*

$$\|J_N \mathbf{x}_n - f_n\|_{\mathcal{H}} \leq \kappa^n \|J_N \mathbf{x}_0 - f_0\|_{\mathcal{H}} + \varepsilon_N \frac{1-\kappa^n}{1-\kappa}.$$

The convention $\kappa^0 = 1$ applies also at $\kappa = 0$. If $\varepsilon_N \to 0$, the transition maps converge uniformly on C.

Proof. Since $J_N^\dagger J_N = \mathrm{Id}$, $J_N \mathbf{x}_{n+1} = F_{N,u}(J_N \mathbf{x}_n)$. Insert $T_u(J_N \mathbf{x}_n)$, apply the uniform discrepancy and contractivity of T_u, and obtain $d_{n+1} \leq \varepsilon_N + \kappa d_n$. Induction gives the geometric-sum bound. Uniform convergence is the definition of the discrepancy bound with $\varepsilon_N \to 0$. □

Corollary 40.13 (Convergence of Fixed Points)**.** *Let f_u^* and $\widehat{f}^*_{N,u}$ be the unique fixed points of T_u and $\widehat{T}^{e,\phi}_{N,u}$. Under the assumptions of Theorem 40.12,*

$$\left\| J_N \widehat{f}^*_{N,u} - f_u^* \right\|_{\mathcal{H}} \leq \frac{\varepsilon_N}{1-\kappa}.$$

Consequently, for a fixed $\kappa < 1$, if $\varepsilon_N \to 0$, then

$$\lim_{N\to\infty} \left\| J_N \widehat{f}^*_{N,u} - f_u^* \right\|_{\mathcal{H}} = 0.$$

Proof. Invariance of C and completeness give $f_u^* \in C$. Write $g_N = J_N \widehat{f}^*_{N,u} \in C$. Then $F_{N,u}(g_N) = g_N$ and

$$\|g_N - f_u^*\| \leq \varepsilon_N + \kappa \|g_N - f_u^*\|.$$

Rearrange. No uniform gap for the discrete contraction constants ρ_N is needed for this estimate; each finite fixed point must exist, and the continuum factor κ supplies the denominator. □

Lemma 40.14 (Cell-Average Error for a Lipschitz Profile)**.** *Let $a : [0,1] \to \mathbb{R}$ be L_a-Lipschitz and put $P_N = J_N J_N^\dagger$. Then*

$$\|P_N a - a\|_{L^2} \leq \frac{L_a}{\sqrt{12}\, N}.$$

Proof. For a cell I of length $h = 1/N$ and its mean $\bar{a}_I$, the variance identity gives

$$\int_I |a - \bar{a}_I|^2 = \frac{1}{2h} \int_I \int_I |a(s) - a(t)|^2 \, ds\, dt \leq \frac{L_a^2 h^3}{12}.$$

Sum over N cells and take square roots. For an affine profile the bound is attained with L_a equal to its absolute slope. □

Theorem 40.15 (Realized Contractive Bridge with Nonconstant Equilibria)**.** *Let $0 < \kappa < 1$, let $\mathcal{U}$ be a nonempty compact interval, and let a nonconstant Lipschitz profile $a : [0,1] \to \mathbb{R}$ satisfy*

$$0 \leq a(s) + u \leq 1 - \kappa \qquad (s \in [0,1],\ u \in \mathcal{U}).$$

Take $\mathcal{I} = C$, $Kf = \kappa \int_0^1 f$, and $\Gamma(u)(s) = a(s) + u$. Let $a_{N,i} = N \int_{I_{N,i}} a$ and

$$\widehat{T}_{N,u}(\mathbf{x})_i = \mathrm{clamp}_{[0,1]} \left(\kappa \frac{1}{N} \sum_j x_j + a_{N,i} + u \right).$$

This finite transition has the original increment form: choose all N nodes as incoming neighbors (including self-edges), $I_j(x_j) = x_j$, $e_i = \phi_i = 0$, $k = T = 1$, $\theta = N/\kappa$, $\eta = \alpha = \beta = 1$, $D = 0$,

$B(\mathbf{x}) = -\mathbf{x}$ *and* $\Gamma_{N,i} = a_{N,i} + u$. *The assumptions of the discrete contraction and bridge theorems hold with* $\rho_N = \kappa$ *and*

$$\varepsilon_N = \|P_N a - a\| \le \frac{L_a}{\sqrt{12}\,N}.$$

Writing $\bar{a} = \int_0^1 a$, *the equilibria are*

$$f_u^*(s) = a(s) + u + \kappa\frac{\bar{a}+u}{1-\kappa}, \qquad \widehat{f}^*_{N,u,i} = a_{N,i} + u + \kappa\frac{\bar{a}+u}{1-\kappa}.$$

In particular, H1--H5 hold and

$$\left\|J_N \widehat{f}^*_{N,u} - f_u^*\right\| = \|P_N a - a\| \le \frac{L_a}{\sqrt{12}\,N}, \qquad \operatorname{dist}(f_u^*, \mathcal{N}) = \|a - \bar{a}\| > 0.$$

These conclusions hold simultaneously for every $u \in \mathcal{U}$.

Proof. C is nonempty, closed and convex in L^2, $\|K\| = \kappa$, and $u \mapsto \Gamma(u)$ is Lipschitz on compact $\mathcal{U}$. The range condition makes the clamp inactive on C and on C_N. The chosen graph has energy--phase coefficient $1/\theta = \kappa/N$ on each edge, so the increment $\mathbf{x} + R(\mathbf{x}) + B(\mathbf{x})$ equals the constant vector $\kappa N^{-1}\sum_j x_j$. The contraction factor is therefore κ in both spaces.

Cell averaging preserves the mean. Consequently $F_{N,u}(f) - T_u(f) = P_N a - a$ for every $f \in C$, proving the uniform discrepancy and its rate by Lemma 40.14. Averaging either fixed-point equation gives $m = \kappa m + \bar{a} + u$, hence $m = (\bar{a}+u)/(1-\kappa)$ and the displayed formulas. Their difference is exactly $P_N a - a$. The orthogonal projection of f_u^* onto the constant subspace is its mean, leaving $a - \bar{a}$ as residual. A nonconstant continuous profile has positive residual norm. Thus no equilibrium is constant, and uniqueness excludes a constant fixed point: H5 follows for this class rather than being an additional assumption about it. □

Example 40.16 (Explicit Common Witness). Choose $\kappa = 1/4$, $a(s) = s/4$ and $\mathcal{U} = [0, 1/4]$. All preceding conditions hold. Directly,

$$f_u^*(s) = \frac{s}{4} + \frac{1}{24} + \frac{4u}{3}, \qquad \left\|J_N \widehat{f}^*_{N,u} - f_u^*\right\| = \frac{1}{4\sqrt{12}\,N}, \qquad \operatorname{dist}(f_u^*, \mathcal{N}) = \frac{1}{4\sqrt{12}}.$$

For $N \ge 2$ the finite equilibrium coordinates are also nonconstant. This is a mathematical realization of the projection dynamics, dissipative finite transition, consistency and non-collapse obligations together. It does not establish the inverse-limit non-finite-determination condition, the six operational invariants or a phenomenal bridge.

41 Projective Dissipation on the Circle-Prefix Towers

Fix the two towers of Theorem 15.18, with $w_t = 3^{-t}$, jump a at stage k, and branch $s \in \{0, 1\}$. Write $\beta_k = 3^{-(k+1)}$. A history is specified by an initial angle θ and an infinite bit sequence b, with stage-t angle $\theta + sa\mathbf{1}_{t>k}$. The depth t is distinct from the iteration index n. For a fixed parameter $u \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$, put

$$R(b)_1 = 0, \qquad R(b)_{j+1} = 1 - b_j, \qquad D_{s,u,t}(\theta, b_1, \dots, b_t) = \big(u + sa\mathbf{1}_{t>k},\, R(b)|_t\big).$$

Here $R(b)|_0$ is empty, and the right side at depth t depends only on the given prefix. Write $D_{s,u}$ for the corresponding evolution of complete histories.

Theorem 41.1 (Contractive Projective Evolution and Equilibrium Transport)**.** *The stage maps commute with their bonding maps and with the existing transport* h_t *between towers. At fixed* u,

the map $D_{s,u}$ has sharp Lipschitz constant $1/3$ on I_s in d_w. Its unique fixed history $x^*_{s,u}$ has initial angle u and binary sequence $0101\ldots$. For $n \geq 0$ its orbit satisfies

$$d_w(D^n_{s,u}x, x^*_{s,u}) \leq 3^{-n} d_w(x, x^*_{s,u}).$$

The existing transport H conjugates the two evolutions and sends $x^*_{0,u}$ to $x^*_{1,u}$. Their d_w and common readout distances are both $\beta_k a$. The finite readouts retain E0--E2 on each history domain, with the original uniform tail 3^{-t}.

Proof. Deleting the last bit commutes with taking the stated initial segment of $R(b)$. At stage k, subtracting sa from the updated phase $u + sa$ gives u as required. At every other stage the updated phase is the same on both sides of the bonding map. Transport adds a precisely at $t > k$ and leaves the bits fixed, giving the transport square. The induced history equations follow coordinatewise.

For points in one tower the phase difference vanishes after one iteration. If the input bits first differ at m, the output bits first differ at $m + 1$. The binary distance therefore changes from 3^{-m} to $3^{-(m+1)}$. Equality in the contraction bound is attained with equal input phases and distinct bits. The fixed bit sequence is forced by $b_1 = 0$ and $b_{j+1} = 1 - b_j$, and the fixed angle by the phase update. Iteration gives the claimed geometric estimate. The phase difference under H is a at levels $t > k$, and Theorem 15.18 gives the readout distance. The underlying domains and readout are the same as in that theorem. A later bit may still be flipped within each finite cylinder, which proves E2 on the history domain. □

Theorem 41.2 (Hilbert Realization of the Projective Dissipation)**.** *The circle-prefix evolutions above embed into the BaseCore projected dynamics on $L^2([0,1])$ with $C = \{0 \leq f \leq 1 \text{ a.e.}\}$, $\|K\| = 1/3$, and compact parameter space $\{0,1\} \times \mathbb{T}$. The embedding is bi-Lipschitz on $I_0 \cup I_1$ for fixed k. The equilibrium profiles $f^*_{s,u}$ satisfy*

$$\inf_{s,u} \operatorname{dist}(f^*_{s,u}, \mathcal{N}) \geq \sqrt{2/45},$$

where $\mathcal{N}$ is the subspace of constant L^2 functions. Thus H5 follows for this family.

At depth t let Q_t be the orthogonal projection onto the retained profile cells, with zero tail. The realized finite map has $N_t = 8 \cdot 9^t$ equal-mesh nodes and its embedded update is $Q_t T_{s,u}$. It commutes with restriction in the sense $Q_t T_{s,u} = Q_t T_{s,u} Q_t$ on C. Its uniform one-step discrepancy on C is at most $3^{-t}/\sqrt{2}$, with equality for $t \geq 1$. Its fixed point is $Q_t f^*_{s,u}$, with error at most the same bound. Starting the finite evolution at $Q_t f_0$ gives this bound at every iteration $n \geq 1$ relative to the orbit from f_0.

Proof. Partition $[0,1]$ up to null sets into four consecutive phase cells of length $1/8$ and consecutive binary cells B_j of lengths $\ell_j = 4 \cdot 9^{-j}$ for $j \geq 1$. These lengths sum to one. Encode each of θ_0, θ_{k+1} by its pair

$$e(\theta) = \big((1 + \cos(2\pi\theta))/2,\ (1 + \sin(2\pi\theta))/2\big)$$

on its two phase cells. Assign b_j on B_j. The chord distance of the encoded angle pair is $\sin(\pi\rho)$ for circle distance $\rho \leq 1/2$, hence lies between 2ρ and $\pi\rho$. At first differing bit m, the norm of the binary part lies between $2 \cdot 3^{-m}$ and $(3/\sqrt{2})3^{-m}$. Since d_w on the union is the maximum of the initial angle distance, β_k times the later angle distance, and the binary distance, these bounds give the claimed bi-Lipschitz property.

For $f \in L^2$, let Kf be zero on the phase cells and B_1, and equal to the negative mean of f on B_j throughout B_{j+1}. Cauchy--Schwarz on each cell and $\ell_{j+1}/\ell_j = 1/9$ give

$$\|Kf\|^2 \leq \tfrac{1}{9} \sum_{j \geq 1} \int_{B_j} |f|^2 \leq \tfrac{1}{9} \|f\|^2.$$

A nonzero function supported on and constant on B_1 attains equality. The forcing $\Gamma(s,u)$ assigns the phase values $e(u), e(u+sa)$, zero on B_1, and one on each later B_j. This is continuous on the compact parameter space. On C, the affine output lies in C, so projection onto C leaves it unchanged. On the embedded histories, it produces exactly the two phase resets and the bit update, preserving each embedded tower for its branch parameter. H1--H4 hold. The unique fixed profile has value zero on B_1 and one on B_2. For any constant c, its squared distance to that constant is at least $\ell_1 c^2 + \ell_2(1-c)^2 \geq \ell_1\ell_2/(\ell_1+\ell_2) = 2/45$. This proves the bound and H5. The same two retained cells certify the lower bound in the finite embedding for $t \geq 2$.

The retained cells have endpoints on the equal mesh of size $1/N_t$. The equal-mesh averaging of f therefore preserves each retained cell mean. To implement the update on mesh nodes, take the resonance term zero and use the feedback $B(\mathbf{x}) = -\mathbf{x} + L_t\mathbf{x}$, with $\alpha = 1$, where L_t is the equal-mesh restriction of Q_tK to piecewise constant inputs. The input is the mesh vector for $Q_t\Gamma(s,u)$. The increment $\mathbf{x} + B(\mathbf{x})$ is $L_t\mathbf{x}$, whose normalized Euclidean operator norm is at most $1/3$ by the mesh isometry. The clamp is inactive on C and the embedded update is $Q_tT_{s,u}$. Every retained output depends only on retained means or the parameter. Hence $Q_tT_{s,u} = Q_tT_{s,u}Q_t$ on C.

The difference $T_{s,u}f - Q_tT_{s,u}f$ is a tail supported on cells of total measure $\frac{1}{2}\,9^{-t}$, with values in $[0,1]$. All tail values equal one for $f = 0$ if $t \geq 1$. For $t = 0$, B_1 is forced to zero and the sharp discrepancy is $1/(3\sqrt{2})$, still within the stated upper bound $1/\sqrt{2}$. The projected fixed point solves the finite equation by commutation, and contraction ensures uniqueness. By induction the finite trajectory from Q_tf_0 equals $Q_tT_{s,u}^n f_0$. The same tail bound applies to $T_{s,u}^n f_0$ for every $n \geq 1$. □

Remark 41.3 (Controls for Projective Dissipation). Holding bits fixed preserves positive distances on equal-phase fibers, ruling out strict contraction in d_w. Replacing the complement by a plain right shift with leading zero keeps the contraction but changes the fixed sequence to $0000\ldots$. At $u = 0$ on I_0 this recovers the original null readout $(0,0,0)$. With the complement, the fixed ternary readout is $\sum_{r\geq 1} 2\cdot 3^{-2r} = 1/4$, so each fixed readout is at least $1/4$ away from that null element. These are separate from H5, whose distance is measured in L^2. For $a > 0$ the branch-preserving map on the union has two distinct fixed points and therefore cannot be a strict contraction there. Although E2 holds on the history domain, the restriction to a singleton equilibrium at a fixed parameter factors through every stage. The model dissipates dependence on the initial history, and its phase component resets in one step.

Example 41.4 (Contractivity Alone Does Not Give Uniform Discretization). Let $T(f) = \kappa f$ on C, with $0 < \kappa < 1$, and $F_N(f) = \kappa P_N f$. In each cell choose f_N equal to 0 on the first half and 1 on the second. Then $P_N f_N = 1/2$ and

$$\|F_N(f_N) - T(f_N)\| = \kappa/2$$

for every N. Thus even strict contractions need additional structure for a uniform approximation rate over all of C. The profile-based realization above supplies that structure explicitly.

Remark 41.5 (Applicability to the Current QICN Runtime). Application to any implementation requires identifying its actual state domain, coefficients, feedback, frozen or evolving auxiliary coordinates, and approximation error. Network size or mean degree alone cannot discharge these conditions. The realized family above is an explicit mathematical example; no assertion is made here that a current runtime implements it or satisfies its hypotheses.

A Counterexamples for Necessity

A.1 Without H1 (Convexity)

Let $\mathcal{I} = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ (unit circle, not convex).

For $\Psi = (0,0)$, every point on $\mathcal{I}$ is equidistant. Projection is not unique.

A.2 Without H2 ($\|K\| < 1$)

Take $\mathcal{H} = \mathcal{I} = \mathbb{R}^2$, $\mathcal{U} = \{0\}$, $\Gamma(0) = 0$ and let K be rotation by $\pi/2$. The point 0 is fixed, but every nonzero orbit cycles with period four and fails to converge. Thus removing strict contraction can destroy convergence even when a unique fixed point remains.

To obtain failure of existence as well, take $K = \mathrm{Id}$ and $\Gamma(0) = v \neq 0$ on the same space. The equation $x = x + v$ has no solution. Both examples preserve H1, H3 and H4.

A.3 Without H3 (Completeness)

H1 already specifies a Hilbert space, so H3 cannot be removed independently while keeping H1 unchanged. To test the extension to a real inner-product space without completeness, take c_{00}, the real finite-support sequences with the ℓ^2 norm, and $\mathcal{I} = c_{00}$. Let $S(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$, $K = S/2$, $\mathcal{U} = \{0\}$ and $\Gamma(0) = e_1$. Projection onto $\mathcal{I}$ is the identity and $T(x) = Sx/2 + e_1$ is a strict contraction. Its only possible fixed-point coordinates are $(1, 1/2, 1/4, \ldots)$, which do not belong to c_{00}. The iteration from zero is Cauchy but has no limit in the stated space. This isolates the completeness requirement for the generalized contraction argument.

A.4 Without H4 (Compactness)

Take $\mathcal{H} = \mathcal{I} = \mathbb{R}$, $K = 0$, $\mathcal{U} = \mathbb{R}$ and $\Gamma(u) = u$. The input map is uniformly continuous, but the parameter space is not compact. Each transition has the unique fixed point $f_u^* = u$, and $\mathcal{A}^* = \mathbb{R}$ is not compact. Thus compactness of the parameter space cannot simply be dropped while retaining the stated compact-image conclusion. It is sufficient, not necessary, since some maps on noncompact parameter spaces have compact image.

A.5 Why Φ-Regularity Is a Hypothesis

Let $\mathcal{I} = \mathbb{R}$ with the Euclidean metric, let $\mathcal{E} = (-\pi/2, \pi/2)$ with the Euclidean metric, and define

$$\Phi(x) = \arctan(x).$$

The map Φ is proper as a map into the open interval $\mathcal{E}$, locally Lipschitz, and non-constant on every nonempty ball. However, no constant $C > 0$ can satisfy

$$|\arctan(x) - \arctan(y)| \geq C|x - y| \qquad \forall x, y \in \mathbb{R}.$$

Indeed, with $y = x + 1$,

$$|\arctan(x + 1) - \arctan(x)| \to 0 \quad \text{as } x \to \infty,$$

while $|x - y| = 1$. Thus properness, local Lipschitz regularity, and non-collapse do not imply a global lower Lipschitz bound. Any theorem using such a bound must carry it as an explicit hypothesis or prove stronger coercive hypotheses.

The same obstruction persists with a complete codomain: let $\mathcal{I} = \mathcal{E} = \mathbb{R}$ and $\Phi(x) = x^3$. This is a proper, locally Lipschitz, injective map between complete spaces and is nonconstant on every nonempty ball. If $|x^3 - y^3| \geq C|x - y|$ held globally for $C > 0$, choosing $y = 0$ and $x = h \neq 0$ would imply $h^2 \geq C$ for all h, a contradiction as $h \to 0$. Completeness of the codomain does not supply a positive lower Lipschitz constant.

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